

Sun, Sand, and Services: Tourism and Household Welfare in Jamaica

Matthew McKetty

Department of Agricultural and Applied Economics, University of Wisconsin - Madison

Abstract

Tourism services have seen marked growth over the last two decades, and a considerable portion of this growth has occurred in developing countries. A growing number of lower and middle income countries aim to harness growth in tourism for the purposes of increasing living conditions, lowering poverty, and raising overall prosperity. Despite its importance, there are still questions about tourism's ability to drive inclusive economic development. This paper contributes to the literature by answering this question in the context of Jamaica, an emerging economy that has made the export of tourism services the foundation of its national development strategy. I combine a rich repeated cross-section of Jamaican household expenditure and a granular tourist spending data from the Jamaican Ministry of Tourism spanning the years 2001-2021. I then employ a shift-share instrumental variable identification strategy to estimate the relationship between tourism revenues and household well-being. I find that an increase in tourist accommodation spending of 10 million US dollars yields a 2.7 percent increase in real per-capita expenditures for urban households. I find that these increases in expenditure occur across areas of food and nonfood consumption, along with healthcare. These increases are concentrated in medium-skilled occupations in non-tourism service sector and manufacturing industries, and for households roughly in the middle of the expenditure distribution.

Introduction

The growth of service sector industries has become a defining feature of structural change in modern developing economies. Tourism has been one of the most important sources of growth within global services ([World Tourism Organization 2023](#)), and this has been especially true for lower and middle income countries (LMICs) ([Nayyar et al. 2021](#)). Despite the undeniable growth in tourism services, there remain significant questions about the ability of tourism to generate economic growth that raises real consumption levels across a broad cross section of the population. There is also uncertainty about the scope for tourism to generate positive spillovers to other sectors of the economy through forward and backward linkages.

Some of the existing work on tourism services has found that tourism-specializing regions experience long-run economic benefits in part driven by spillovers to manufacturing ([Faber and Gaubert 2019](#)). Other work focused on the short-run effects of urban tourism in large cities has found that benefits to local residents in terms of income or improved amenities must be considered against cost of living increases and the extent to which local amenity preferences align with those of tourists ([Almagro and Domínguez-Iino 2025](#); [Allen et al. 2021](#)). Much of the existing work has either taken place in industrialized countries with large domestic tourism industries, or has focused on the long-run effects of tourism in a developing country context. There is a lack of evidence from developing economies for tourism’s short-run effects on local households of varying skill and income levels, partially owing to data constraints and partially owing to the difficulty of measuring tourism activity. This paper contributes to the literature by investigating the relationship between tourism and household welfare, quantifying its effects on different skill segments of the labor force, and characterizing the heterogeneity of its impacts across socioeconomic segments in Jamaica, an upper-middle income country that has made tourism a centerpiece of its development strategy.

In this study, I answer two questions: Do increases in tourist in Jamaican localities increase real per-capita consumption expenditures for households within that locality? And conditional on observing increases in real per-capita consumption, to what extent are these consumption increases observed among households below or near the poverty line. More broadly, this paper sets out to provide answers to the larger question of whether or not it is possible to leverage tourism for major economic development that raises living standards throughout a local population, particularly for the poorest. I show that in the short-run, increases in an area’s tourism revenues results in an increase in real consumption levels, lowers poverty, and primarily benefits those in the lower quartile of the consumption distribution, but only for urban households. This paper harnesses two unique and granular datasets, one a detailed survey of tourist spending behavior and the second, nationally representative household expenditure surveys, both for the period 2001 – 2021. In order to account for the likely endogeneity of local tourism in different Jamaican communities, I employ a shift-share instrumental variable strategy, exploiting exogenous variation in the number of tourists from different regions of origin who choose to visit Jamaica in a given year, as well as variation in their decisions of where to stay in Jamaica during their vacations. There was significant temporal and spatial variation in tourism activity and household welfare across the country during the period 2001-2021. Those years saw marked growth in the industry, but

also considerable volatility. This variation enables me to identify the effects of tourism growth on locals.

The growth of tourism Jamaica, the heterogeneous exposure of different communities across the island to the industry, and sizable year-to-year variation in industry performance motivates the study of the effects of the tourism sector on local communities as demand- shocks for tourism services in a particular locality. Tourism’s labor intensity means that an increase in demand for tourism services can be expected to produce an increase in demand for local labor. As a consequence of its debatable impact on local communities and its tradable nature, tourism has also drawn comparisons to natural resource ventures. To guide this analysis I draw from the predictions of work on tourism (Allen et al. 2021; Almagro and Domínguez-Iino 2025), as well as work on local demand shocks and resource booms (Moretti 2010; Aragón and Rud 2013). The framework of local labor demand shocks of Moretti (2010) predicts that an increase in demand for labor-intensive services such as tourism will raise local earnings (and therefore local consumption levels), relative to earnings in communities that do not experience the shock. This may then result in increases in earnings for workers in other sectors through increased consumption levels. A positive tourism shock may also produce inflationary effects because of the increase in local earnings as a result of the labor demand shock (Moretti 2010; Aragón and Rud 2013), or through tourists consuming local goods and services (Allen et al. 2021; Almagro and Domínguez-Iino 2025). The skill levels of the workers who benefit are predicted by Aragón and Rud (2013) to depend on the skill-composition of the industry experiencing the demand shock. Given that lower to mid-skilled occupations comprise most of the Jamaican tourism workforce, it is reasonable to expect a tourism shock to yield gains for those workers. Finally because those in the lower segments of the expenditure distribution are more likely to have lower to mid-skilled occupations, I would expect households below or near the poverty line to experience the greatest benefit in the Jamaican setting.

I find that greater development area tourism revenues have a positive effect on real per-capita expenditures. In particular, an increase in area revenues of 10 million U.S. Dollars generates a 2.7% increase in per-capita expenditures for urban households within a development area relative to households in districts that did not receive the shock. Because the median tourism districts see increases in real earnings of roughly 7.9 million in a given year, these figures translate to an increase in real expenditures of 97 U.S. Dollars relative to households in areas that do not experience an increase in tourism levels. Households increase their food and non-food expenditures by 2.3% and 2.9%, respectively, though these are not statistically different from one another. Increased real expenditure on medical services and products represents 33% of the change in non-food expenditures, while there is no change in real spending on education or loan repayments. Lending further support to the interpretation of these effects as improvements in well-being, households are .6 percent less likely to be below the poverty line for every 10 million dollar increase in tourism revenues.

As expected, there is an increased likelihood that a surveyed household works in a tourism related industry following an increase in development area tourism revenues. Surprisingly, the increases in real expenditures are driven by those who work in non-tourism sectors, who on see an increase in real per-capita spending by of 2.9 percent per additional 10 million U.S. Dollars in tourism revenues. There is no statistically signifi-

cant effect on real expenditures for households working in the tourism sector. Similarly, decreases in poverty are entirely driven by non-tourism households. I subsequently show that it is urban skill-level 2 occupations (those requiring up to a secondary education) that are the primary beneficiaries of increased tourism levels. The level 2 occupations that benefit are “Service, Craft, and Trade Work” and “Plant and Machine Operators”. Using a quantile regression, I find that urban households in the bottom two quartiles of the urban consumption distribution who benefit from increased tourism levels. When compared against all Jamaicans, these households are in the 2nd and 3rd quartiles, indicating that on average the impacts are primarily impacting the middle and lower-middle segments of the expenditure distribution.

Finally, I provide additional support for these results by utilizing an unbalanced panel of households that is a subset of the main repeated cross-sections. The panel regressions show that effects of increases in tourism revenues are concentrated among the households that are employed in non-tourism services and manufacturing during all periods that they are observed, and who are in quartiles 1-3 of the consumption distribution in the first period they are observed. In order to confirm that the results I observe are not driven by cost of living increases I run two additional regressions. The first shows that there is no effect of tourism earnings on real rent expenditures for households. A final regression shows that the effects of tourism are driven by workers in private sector employment with no change in real consumption for government workers whose salaries are set nationally and are unlikely to reflect changes in local tourism levels, thus further lending support that the observed increases in expenditures reflect increases in real consumption and welfare.

Jamaica is an informative setting in which to study the relationship between tourism and local household welfare for two reasons. The first is that Jamaica is comparable with a key subset of countries in its level of economic development and industrialization, the type of tourism in which it has a comparative advantage, and the skill levels of its labor force. Jamaica is a middle-income country heavily dependent on services, which make up 70% of its economic output. This is similar to the large group of developing economies increasingly specializing in services and failing to industrialize ([Rodrik 2016](#)). An island nation located in the Caribbean Sea and endowed with world famous beaches and a tropical climate, Jamaica has specialized in “Sun, Sand, and Sea” style tourism that is most popular in such climates. Regions endowed with similar types of climates include Latin America, Sub-Saharan Africa, Southeast Asia, and the South Pacific. As is the case for many middle-income countries, Jamaica’s labor force is overwhelmingly made up of workers qualified for low and mid-skilled occupations.

A primary development aim for emerging economies is to develop industries that are able to absorb this large supply of labor, and this is one of the major strengths of tourism ([Nayyar et al. 2021](#)). The second attribute of Jamaica that makes it useful for study is the country’s aggressive pursuit of tourism lead growth. Jamaica’s shared characteristics with a number of countries and its pursuit of tourism oriented development mean that findings from this setting can provide insights relevant to a number of other countries. These findings may be useful for countries that have also pursued tourism lead growth and wish to better understand their outcomes, or they can be informative for places considering such a growth strategy. The interest in tourism

as an engine of development can be seen in the diverse set of countries that have developed strategic plans for expanding their tourism industries. These include: Indonesia¹ and India².

This paper contributes most directly to the literature studying the relationship between tourism and economic development. One strand of this literature attempts to quantify the impacts of tourism specialization through wages, sectoral spillovers, and agglomeration economies. Research by [Faber and Gaubert \(2019\)](#) in Mexico used a quantitative spatial equilibrium model and demonstrate that tourism-specializing municipalities experience significant long-run gains relative to regions that do not specialize in tourism, with backward linkages to manufacturing serving as a key mechanism. Studying the extent to which tourism benefits the poor in Thailand [Wattanakuljarus and Coxhead \(2006\)](#) use a general equilibrium approach and find that growth in tourism raises aggregate income but worsens inequality. Work by [Çiftçi et al. \(2007\)](#) showed that tourism produced short-run gains that did not translate to the long term. Other work in the tourism literature has focused on using urban spatial equilibrium to quantify the short-term welfare impacts of tourism in modern cities. In Barcelona, [Allen et al. \(2021\)](#) combine reduced-form analysis with general equilibrium modeling to show that tourism’s effects on local welfare exhibits a tension between tourist-induced cost of living increases and tourism-driven wage growth. In Amsterdam, [Almagro and Domínguez-Iino \(2025\)](#) detail how the growth of short-term rental tourism impacts the development of residential amenities and further describe how the welfare impacts of these changes are a function of the distribution of local preferences and demographic characteristics over amenities.

This study complements these existing works on several dimensions. Whereas [Faber and Gaubert \(2019\)](#) and [Wattanakuljarus and Coxhead \(2006\)](#) focus on long run municipal and national level outcomes, this paper exploits the temporal and demographic granularity of the data to quantify the heterogeneous short-run effects of tourism sector growth. The richness of the data allows for reduced form evaluation of the general equilibrium channels modeled by [Faber and Gaubert \(2019\)](#) and [Wattanakuljarus and Coxhead \(2006\)](#). This work also contributes by studying tourism services in a country where they are an overwhelmingly exported product, with more than 90% of Jamaican tourism revenue coming from international visitors in 2023 ([STATIN 2019](#)) compared to approximately 20% in Mexico ([Faber and Gaubert 2019](#)). To the extent that the export share of tourism is associated with different skill requirements for workers, interactions with the local economy, and vulnerability to external shocks, this distinction could have important welfare implications for tourism on the welfare of local residents.

This study also contributes to the expansive literature that works at the intersection of service-led structural transformation and economic development. Canonical models such as the Dual Sector or Lewis Model ([Lewis 1954](#)), postulate that the transition of surplus labor out of agriculture, and into high productivity manufacturing enables the gains in productivity that generate economic growth, and raise the living standards of a population. However, in recent years the relative share of manufacturing in many economies has

¹Ministry of Public Works and Housing 2018.

²Indian Ministry of Tourism 2022.

fallen (Rodrik 2016). Work such as that by Nayyar et al. (2021) also view services such as tourism as unlikely to possess the ability of manufacturing to absorb low-skilled labor while also generating productivity growth. Some work, has shown the potential strengths of specializing in services. For example, Fan et al. (2023) show that India gained from service sector specialization in large part because of the increase in productivity in urban consumer services that benefited wealthier urban dwellers. Work by McCullough (2025) in Tanzania shows that service-focused structural transformation has the ability to generate local employment, the scope for long-term growth may be limited due to the constrained size of local markets and low productivity growth. This paper provides insight into the implications of tradable service sector specialization for local residents, and also is able to characterize how the effects of shocks to a tradable service sector impact heterogeneous households.

Other studies have detailed the potential challenges of non-industrialization driven structural change. Studies by Gollin et al. (2016) and Venables (2017) discuss how gains from urbanization depend on whether or not the industries in an urban area exhibit productivity gains. They also point out that urbanization that draws large numbers of people into lower productivity non-tradable services could, in fact, be a drag on longer term gains. In this way such a type of urbanization could contribute to effects consistent with the “Dutch-Disease” framework developed by Corden and Neary (1982), or the “Cost Disease” framework introduced by Baumol and Bowen (1965). This paper provides empirical evidence of how shocks to a major tradable service sector impact household welfare in a country that has largely progressed directly from being based on natural resources, to being based on services.

Finally, this paper contributes to the literature studying the effects of sectoral shocks on local economies. A significant subset of this literature relates to the effects of resource windfalls or natural resource ventures on local communities (Aragón and Rud 2013; Aragón and Winkler 2023; Bonilla Mejía 2020; Boire 2021). Other studies have investigated the effects of manufacturing growth on decisions regarding human capital accumulation in settings such as Bangladesh Heath and Mobarak (2015) and Mexico Atkin (2016). This paper provides novel insights into the magnitude of the effects of tourism booms on the well-being of local households in a developing country. As a major global sector that has experienced considerable and has come to dominate many economies, understanding tourism’s effects on the communities it impacts is of great relevance.

In Section 1, I present background information on the Jamaican economy and tourism. Section 3, details the effects of tourism shocks predicted by the literature and lays out four testable hypotheses. Section 4 explains the datasets used for the study, while Section 5 presents the empirical specification. Section 6 explains the findings of the study and Section 7 discusses the implications of the findings and concludes.

1 Context and Background

1.1 Jamaica and Tourism

Jamaica is an island nation in the Caribbean Sea with a population of 2.8 million people. An upper middle-income country, Jamaica’s economy is heavily reliant on tourism, which directly accounts for over 10% of GDP, and over 30% when considering spillovers (Mooney 2020). The tourism sector also comprises about 30% of the labor force when measuring direct and indirect employment (Mooney 2020). Other Caribbean countries have comparable levels of specialization in tourism services, with tourism comprising an average of 25% of Caribbean GDP between 2015 and 2019 according to the OECD (OECD and Inter-American Development Bank 2024).

Table 1: Global, Regional, and Jamaican Tourism Statistics

Indicator	Value
Panel A: Global & Caribbean Region Tourism Statistics	
International Tourist Arrivals (2023)	1.3 billion USD (UNWTO)
International Tourism Export Revenues (2023)	1.8 trillion USD (UNWTO)
Global tourism GDP share (2023)	3% (UNWTO)
Share of Global Trade in Services	23% (UNWTO)
Average Growth In International Arrivals (2000-2023)	4.1% annually (UNWTO)
Caribbean tourism arrivals (2024)	34.2 Million (CHTA)
Tourism Average Share of Caribbean GDP (2015-2019)	25.4% (OECD)
Panel B: Jamaican & Caribbean Tourism	
Total Tourist Arrivals to Jamaica (2024)	4.15 Million (MOT)
Total Stopover Arrivals to Jamaica (2024)	2.9 Million (MOT)
Total Cruise Passenger Arrivals to Jamaica (2024)	1.25 Million (MOT)
Total Roomnights Sold in Jamaica (2024)	5.75 Million (MOT)
Jamaica Total Tourism Earnings (2024)	4.2 billion USD (MOT)
Panel C: Jamaican Tourism Sector Capital Stock & Infrastructure	
Jamaica Hotel Room Capacity (2024)	26,427 Rooms (MOT)
Average Growth in Hotel Room Stock (2000-2024)	2.6% (MOT)
Total Number of Hotels (2024)	210 Properties (MOT)
Total Number of Workers in Accommodation & Restaurant Services (2024)	43,913 (MOT)
2025/2026 Ministry of Tourism Budget	95.5 Million USD (MOT)

Source: United Nations: World Tourism Organization, Caribbean Hotel & Tourism Association, Jamaican Ministry of Tourism (MOT).

Following independence in 1962 the Jamaican government worked to build out the country’s tourism and mining sectors, exploiting the natural resources of “beaches and Bauxite” as described by King (2001). The government created the Ministry of Tourism and instituted various laws such as The Tourist Board Act (Parliament of Jamaica 1969), and the Hotel Incentives Act (Parliament of Jamaica 1971) to support the development of the new industry. Considerable investments were made in infrastructure, advertising, and in attracting foreign direct investment. The geographic base of the industry is along Jamaica’s northern and western coasts, although in recent years there has been increasing development along the Southwest and Northeast coasts as well. In 2019, over 75% of visitors stayed at accommodations located in either Saint James, Trelawny, Saint Ann, Saint Mary, Westmoreland or Hanover.

Of the more than 4.2 million visitor arrivals in 2024, roughly 2.7 million were “stopover” visitors; those who spent at least 24 hours in the country according to the MOT (Jamaican Ministry of Tourism 2025). The

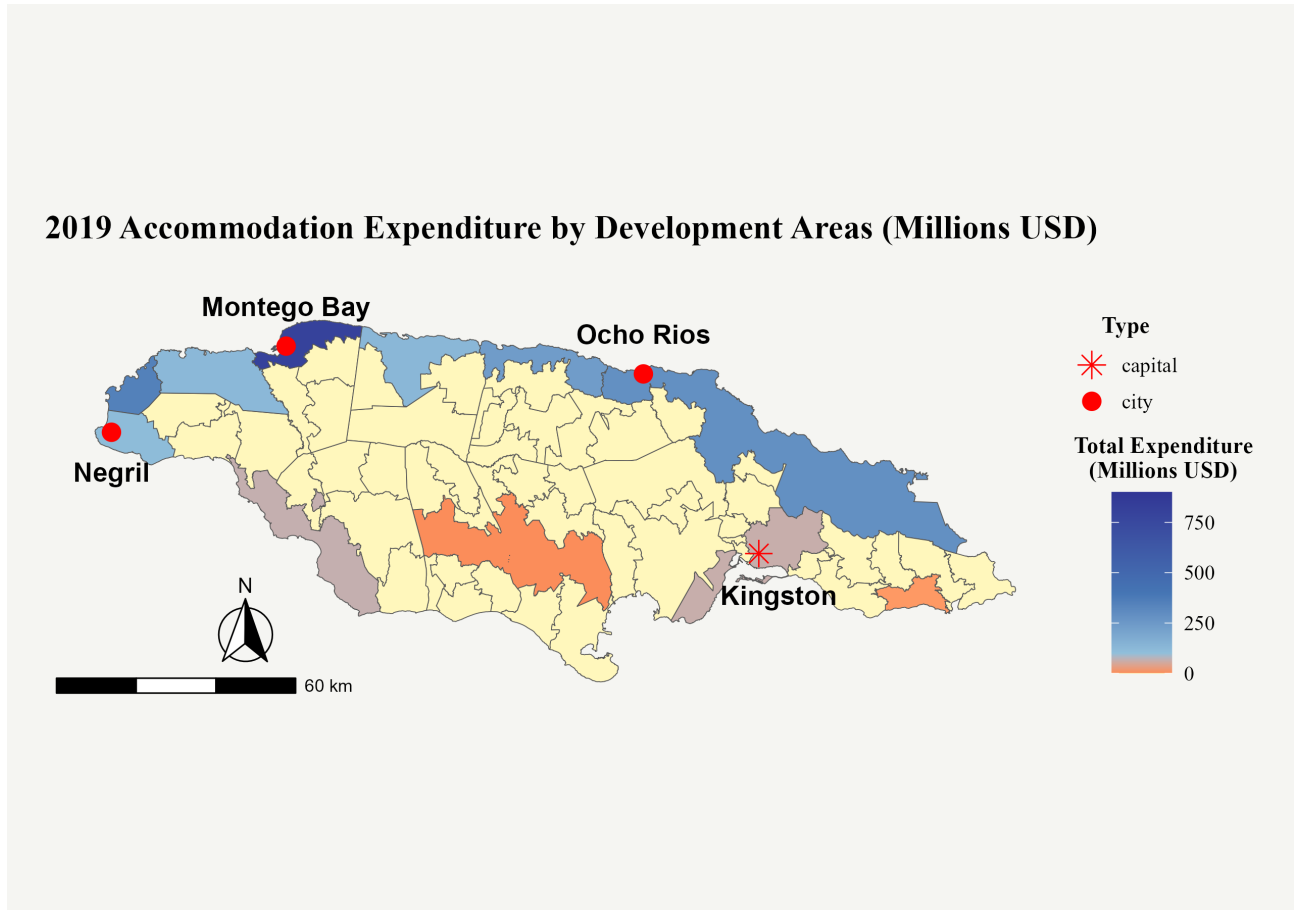


Figure 1: **Tourism Accommodation Expenditures Across The Island**

remaining 1.6 million are cruise visitors, who typically do not spend more than 24 hours in the country. Although stopover visitors made up about 56% of total arrivals to Jamaica, their spending made up more than 95% of total visitor expenditure, as can be seen in the MOT Annual Statistics publication for 2019 ([Jamaican Ministry of Tourism 2020](#)). Therefore, in order to best understand the impact of tourism on economic development, it is reasonable to devote the greatest amount of attention to the stop-over segment of the population of visitor arrivals.

As expected, the tourism industry is an important employer. According to the IADB, the tourism sector employed approximately 250,000 Jamaicans in 2019 either directly or indirectly ([Mooney 2020](#)). This was roughly a quarter of the labor force at the time. Given spatial variation in the intensity of touristic activity and temporal variation in this intensity, there exists the opportunity to causally identify the impacts of tourism on Jamaican households.

The tourism sector has grown considerably over the last 60 years, and since 2000, arrivals have grown on average 5% per year and the room stock has grown at an average annual rate of 2.6%. However, the strong performance of the tourism industry has not translated into similar success in larger economy. Over the same

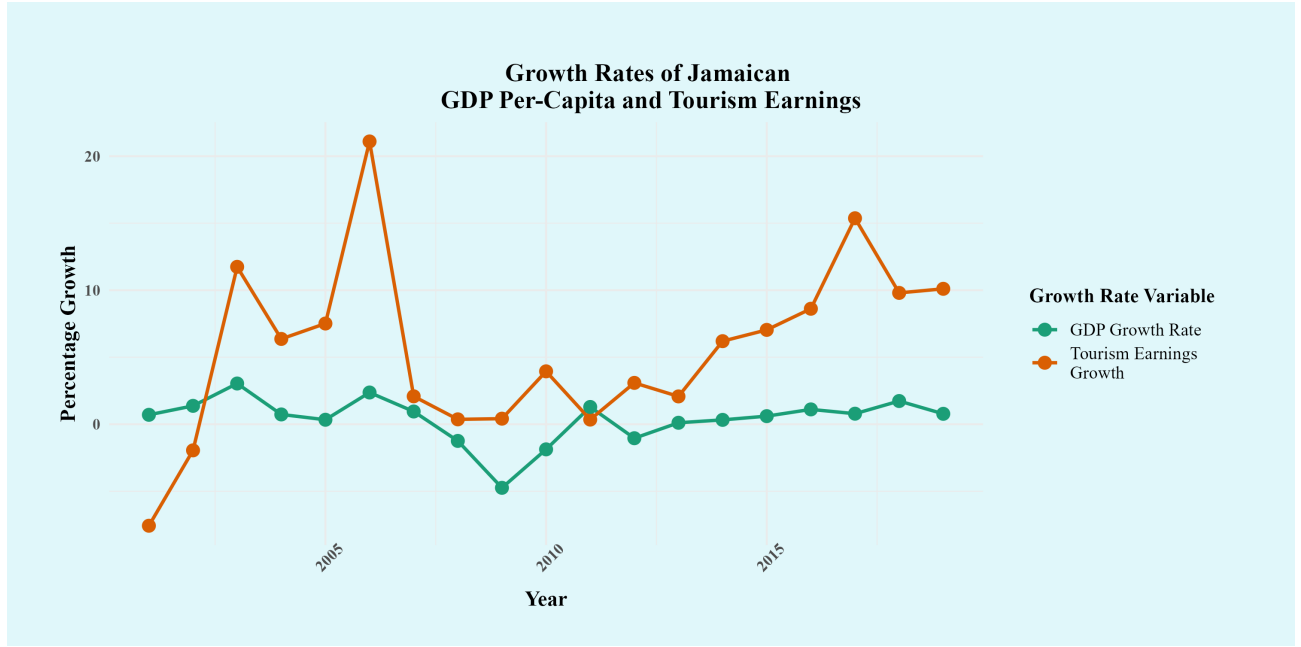


Figure 2: Tourism vs. GDP Growth
Source: World Bank

period, growth rates of real GDP per capita have oscillated between 1 and -1 percent. Value added per worker has been mostly stagnant, even as the tourism share of the labor force has grown and the industry is marketed as the key to generating long-term national prosperity.

The vast majority of tourism activity in Jamaica is much less integrated with the local Jamaican economy than is the case for either Barcelona or Amsterdam. A major reason for this is the dominance of the all inclusive resort model of tourism in most of the country’s tourist areas, as is the case in many developing economies dependent on tourism (Tavares 2015). This resort model is characterized by accommodations that provide a comprehensive package of accommodations, dining and activities, restricted primarily to the resort premises (Issa and Jayawardena 2003). As a result, tourists often spend considerably less in the communities in which resorts are located than when staying in other types of accommodation (Çiftçi et al. 2007). These differences in spending patterns and the structure of tourism may well result in very different welfare outcomes even from shocks of comparable magnitudes, which further supports the relevance of my study.

2 Jamaican Economy Background

2.1 Household and Labor Force Heterogeneity

Despite Jamaica’s relatively small size there is great variation in consumption levels across different regions of the country. Average real consumption expenditure per capita is 3021 USD for households during the study period, real urban consumption expenditures are on average 34 rural consumption expenditures as shown in table 2. There are statistically significant differences in the non-food share of expenditures, which is 4 per-

centage points higher in urban areas relative to rural areas.

Table 2: Household Characteristics: Urban - Rural Comparison

	Urban Households		Rural Households		Comparison
	Mean	SD	Mean	SD	T-Statistic
Per-Capita Consumption	3602.86	3161.80	2682.49	2292.67	25.35***
Per-Capita Total Expenditure	3990.83	3983.195	2889.95	2762.260	24.26***
Per-Capita Food Expenditure	1613.62	1289.520	1355.71	1308.996	14.04***
Per-Capita Non-Food Expenditure	1990.36	2319.748	1327.30	1362.054	26.84***
Per Capita Non Consumption Expenditure	523.55	1487.020	292.83	1008.639	11.60***
Non-Food Share of Consumption Expenditure	0.52	0.145	0.48	0.135	24.67***
Non-Food Share of Tot. Expenditure	0.48	0.139	0.46	0.131	19.51***
Consumption Share of Tot. Expenditure	0.95	0.095	0.96	0.080	-12.44***
Observations	11196		18370		29567

Notes: All statistics are weighted by household size. Panel A reports means, medians, and standard deviations for the full sample. Panel B compares urban and rural households using t-tests with unequal variances. *** p<0.01, ** p<0.05, * p<0.1.

3 Evaluating The Impact of Tourism

Rapid growth has been a key feature of the global tourism industry over the last two decades. The United Nations World Tourism Organization estimates that the number of global travelers has increased from roughly 700 million 25 years ago to 1.3 billion in 2024 ([World Tourism Organization 2024](#)). According to the World Travel and Tourism Council, tourism made up roughly 10% of global GDP in 2019, accounting for 25% of all new jobs created worldwide ([World Tourism Organization 2023](#)). Following the COVID-19 pandemic which saw the collapse of the industry, the global tourism sector rebounded to 80% of pre-pandemic levels in the first quarter of 2023, with roughly 235 million people traveling internationally; more than double the number during the same period in 2022 ([World Tourism Organization 2023](#)).

Tourism has a number of qualities that make it an attractive sector in which to specialize for a developing country. It is an excellent source of foreign exchange and as a labor-intensive sector it is capable of absorbing a large number of unskilled workers ([Nayyar et al. 2021](#)). However, tourism is also characterized by a lack of linkages to other sectors and low scope for productivity growth ([Nayyar et al. 2021](#)), critical characteristics of industries that produce economic growth.

The Jamaican government’s goals for developing the tourism sector are outlined in the “Vision2030” national development plan. The first three goals are that Jamaican tourism is “inclusive and facilitates broad participation by Jamaica”, that the industry has “An adequate workforce within the sector that is skilled, educated and motivated”, and finally that tourism is “A highly integrated sector which can act as a driver for economic development”. A common criticism of tourism in Jamaica and other developing countries is that it is an exploitative industry that has little connection with the local populations ([Çiftçi et al. 2007](#); [Issa and Jayawardena 2003](#); [Issa and Jayawardena 2003](#)). It is therefore useful to understand the conditions under

which positive tourism shocks raise local consumption levels. The criticisms of tourism’s effects on local communities thus share similarities with criticism of natural resource ventures in poor communities.

In this paper, we can consider a positive tourism shock to an area of Jamaica as an increase in the demand for local tourism services. The theoretical framework of [Moretti \(2010\)](#) interprets this as a sudden increase in labor demanded by tourist accommodations, and predict that this produces an increase in employment in the local tourism sector. These studies also predict that there will be increases in earnings for workers in the industry experiencing the demand shock, assuming that labor is mobile between sectors, and that there is imperfect labor mobility between regions. They predict that the extent to which workers in other sectors benefit depends on the same factors and if the sector is tradable or not. Specifically they predict the local non-traded sector will benefit, while the effects are unclear for the tradable sector.

The largest share of the Jamaican labor force, roughly 83 percent, is employed in skill-level 1 or 2 occupations ³. Among tourism related activities⁴, low and mid-skilled occupations make up 82.3 percent ⁵. Given that these occupations do not require more than a secondary level of schooling ([International Labour Office 2012](#)), and comprise up the largest share of occupations across Jamaica’s sectors it is reasonable to expect that Jamaican workers are largely mobile between tourism industries and other segments of the economy. This mobility implies that a positive shock to tourism may result in significant spillovers for a wide swath of the labor force in a particular locality.

What can be said about the geographic mobility of Jamaican workers? As in many developing countries, there has been considerable movement within Jamaica over the previous 2 decades from rural areas in the country’s interior and south to urban centers. This has been driven by the familiar factors of rural poverty and the expectation of better job opportunities in urban centers. A large portion of this migration has been attributed to the development of tourism. According to the 2022 census, the fastest growing Parishes during the period 2011-2022 were St.James, Westmoreland, Hanover, and Trelawny, which are also major hubs of Jamaica’s tourism industry. While this migration has been considerable, it has been on average much slower than the growth of Jamaica’s tourism industry. This suggests that while there is some mobility for Jamaican workers, there still remain significant frictions. One potential friction may be concerns about crime, as well as difficulties with procuring suitable housing as urban areas struggle to update their infrastructure.

What would be the effects of an influx of new residents into a development area in response to a labor demand shock? The new arrivals would generate an increase in demand for local services as well as housing. This may also produce an increase in the prices of local services. Such an increase in the size of the local labor force would also put downward pressure on wages as a result of an increase in the local labor supply. Whether these shocks also raise *real* consumption depend on the relative magnitudes of the changes in nominal earnings and changes in cost of local goods and services ([Moretti 2010](#); [Aragón and Rud 2013](#); [Allen et al.](#)

³Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions

⁴The Statistical Institute of Jamaica defines tourism related activities as sectors that heavily participate in the tourism industry such as Hotels & Accommodations, Transportation and Logistics, etc

⁵Source: Calculations based on 2001-2021 Jamaica Survey of Living Conditions Surveys

2021; Almagro and Domínguez-Iino 2025). In the case of Barcelona, Allen et al. (2021) show that while on average tourism lowers the welfare of Barcelona’s residents because of increases in cost of living, there is significant heterogeneity in which residents have worse outcomes. In the Jamaican setting, owing to tourists being more segmented from the local population compared to urban styles of tourism like in Barcelona, I do not expect a significant effect of tourism on local food prices. I also would not expect there to be a significant short or medium-run effect on rent as a result of an increase in area tourism earnings. Unlike settings like Amsterdam where a large share of tourists stay in short-term rentals in close proximity to residents, in Jamaica (as in other developing country contexts), tourists have overwhelmingly stayed in resorts, hotels, and resort-villas as table 20 shows.

The occupations and skill-levels that benefit are predicted to depend on the skill-makeups of the industries most-impacted by the demand shock. In the Peruvian case, low-skilled workers were the ones to benefit as the Yanacocha mine demanded inputs from labor-intensive low-skill industries (Moretti 2010). Skill level 1 and 2 occupations make up the majority of Jamaica’s tourism workforce and also form a majority of the workforce overall, so in the Jamaican case, it would seem those groups are those that would be predicted to benefit most. What does the literature imply influences which segments of the expenditure distribution will benefit most? Because those in the lowest quartile of the consumption distribution are disproportionately likely to be in the lower 2 skill-level occupations, and because these constitute the majority of positions in the tourism industry, it is reasonable to expect that households in the first quartile would benefit most from increased tourism.

4 Data

The analysis combine tourist expenditure surveys and household expenditure and living condition surveys.

The primary source for data on households comes from the Jamaica Survey of Living Conditions(JSLC). The JSLC is an annual Living Standards and Measurement(LSMS)-style survey conducted on a representative sample of the Jamaican population. It is administered by the Statistical Institute of Jamaica (STATIN). The JSLC uses a two-stage stratified random sampling design. The modules cover a wide range of topics related to household well-being, such as expenditure across different types of consumption, education levels, health, and labor force participation. In most years there are around 2000 households surveyed, which results in around 6000 individuals being included in the sample, or about 0.3% of the Jamaican population. I construct a repeated cross-section of 30680 households, representing 98883 individuals. The data covers the years 2001-2004, 2006, 2008-2011, 2013-2014, 2016-2019 and 2021.

In order to support findings from the baseline repeated cross-section specification, I also construct an unbalanced panel from a subset of the Jamaica Survey of Living Condition data following the methodology used by Handa (2007) in his analysis of households movement across the income distribution. Prior to 2018 the JSLC would occasionally re-sample households from previous years when the same sampling frame was being

employed. On average about half of the households in one year may have been re-sampled. This the case for the years 2002-2003, 2004-2007, 2008-2010, 2013-2014, and 2015-2016. Following the [Handa \(2007\)](#) methodology I am able to construct an unbalanced panel of approximately 6000 Jamaican households representing some 30,000 individuals.

In the main results of this analysis I exclude the capital Kingston, and the two smaller urban centers of May Pen and Mandeville. These municipalities are not major contributors to Jamaica’s main tourism product, together representing 15% of tourist arrivals in 2024. Due to lack of variation in the relative shares of tourists from different origin regions visiting these locales, the shift-share instrumental variable is under-powered when they are included. Thus this analysis is centered on the urban municipalities that produce the lion-share of Jamaican tourism services and the country’s rural regions.

Tourism Exit Surveys

The data on tourism levels comes in part from the Ministry of Tourism (MOT) exit surveys. The MOT administers the survey each month to visitors leaving through either of Jamaica’s two main international airports. While this survey only covers stop-over arrivals, as mentioned in the previous section, stopover arrivals account for the overwhelming share of Jamaica’s tourism revenue. Thus, this survey captures the vast majority of tourist expenditure on the island. For the study period I construct a repeated cross-section with roughly 80,000 travel parties and 150,000 individuals.

The survey asks detailed questions about visitor characteristics, where in Jamaica they stayed, and how much they spent across categories such as accommodations, entertainment, transportation and food. Crucially the survey also asks visitors staying in hotel or hotel-like accommodations for the name of the establishment, as well as their country or state of origin. I can therefore estimate what share of tourists choose to stay in particular areas of the country, the amount of spending on accommodations within these localities, and the relative contribution to local tourism revenues by tourists from different regions of origin. Because the number of parties interviewed varies across years in the survey, and because the number of tourists interviewed does not always scale with the exact number of arrivals, I scale the estimates of local tourist accommodation expenditures by publicly available aggregate tourism statistics published by the Ministry of Tourism.

I make use of administrative shapefiles for the 2001 and 2011 censuses purchased from the Geographic Services Unit (GSU) of STATIN to link touristic activity and households across space. The shapefiles provides me with the boundaries of 86 ‘development areas’ and 5776 enumeration districts into which STATIN divided the country for data collection and analysis. The boundaries for each census are maintained for the following decade until the following census. While for the 2001 census I do not observe development areas, I still observe household enumeration districts. For data before 2011, I overlay 2011 development area boundaries on 2001 enumeration districts in order to obtain consistent spatial units. If an enumeration district is not wholly contained within a particular development area, I assign the households in that district to the development area where the largest share of the district is located. For the purposes of this analysis, I aggregate

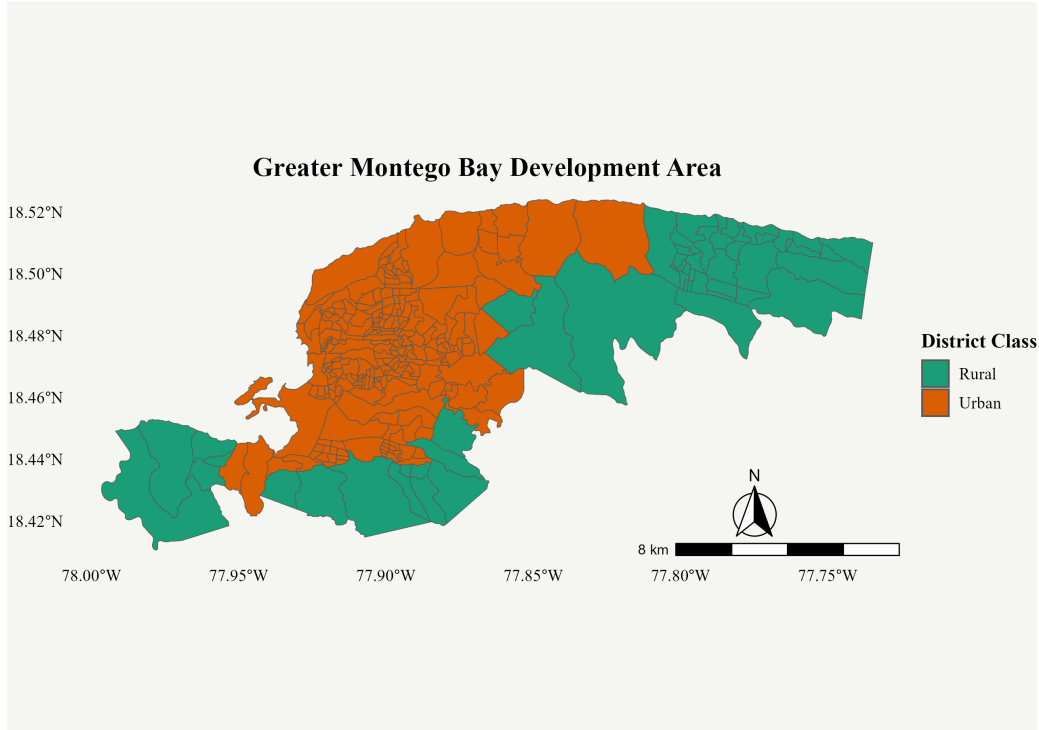


Figure 3: Greater Montego Bay Development Area

Notes: Urban enumeration districts are shaded in orange, while rural enumeration districts are shaded in green.

some neighboring development areas into one.

The final analysis has 65 development areas. Development area boundaries are designed to encompass municipalities with similar economic and social characteristics. Therefore, they represent a useful level at which to estimate variation in tourism activity. They are large enough to contain both urban and rural households as well, as can be seen in figure 3, showing the Greater Montego Bay development area and its urban and rural components.

5 Empirical Approach

This study joins an extensive set of research that employs shift-share instrumental variable (SSIV) identification strategies to correct for endogeneity. Since being first introduced in [Bartik \(1991\)](#) in a study of the growth rate of employment in specific sectors, SSIVs have proven extremely versatile in a variety of areas, including trade and labor ([Autor et al. 2013](#); [Hummels et al. 2014](#)), health ([Miguel and Kremer 2004](#)), migration and labor market outcomes ([Card 2009](#)), and the welfare impacts of location classification ([Diamond 2016](#)).

In constructing and evaluating the SSIV I adapt the strategy of [Allen et al. \(2021\)](#), in the study, exploiting variation in which tourists from certain regions of origin choose to stay in Jamaica and how much they spend. The identification strategy of this paper follows the exogenous shock-based approach [Borusyak et al.](#)

(2022), as opposed to the exogenous share-based approach (Goldsmith-Pinkham et al. 2020). Several papers in recent years have suggested detailed guidelines to ensure accurate shift-share inference in the presence of exogenous shocks, and specifically for this case a panel of exogenous shocks (Borusyak et al. 2022; Borusyak et al. 2024; Borusyak et al. 2024; Adão et al. 2019). Regarding accurate calculation of standard errors in SSIV designs, Borusyak et al. (2024) and Adão et al. (2019) develop techniques to calculate standard errors that account for the correlation in the residuals of units exposed to similar shocks. In their paper, Adão et al. (2019) demonstrate that failure to account for residual correlations can lead to over-rejection of the null hypothesis. I utilize their approach in order to adjust the standard errors for correlation in the shocks and include these regressions in appendix 8.1.

The structural equation describing the relationship that we wish to estimate is given below. In this equation, y_{it} represents per-capita expenditures for household i , in development area d in year t . The term $Tourism_{dt}$ represents total tourist accommodation expenditures in the development area d for the same period, X_{idt} is a vector of household characteristics, and α is a constant and ϵ_{idt} is an idiosyncratic error term. The equation is the following:

$$Y_{idt} = \alpha + \beta Tourism_{dt} + \psi X_{idt} + \gamma_t + \lambda_d + \epsilon_{idt}. \quad (1)$$

The earlier predictions from the literature serve as useful guides for the empirically analyzing the effects of tourism on household welfare for Jamaican communities. The characteristics of the Jamaican tourism sector and labor force, and insights from studies analyzing the effects of local economic shocks (Moretti 2010; Allen et al. 2021; Aragón and Rud 2013), inform four testable hypotheses for this analysis.

- H1: Increases in development area tourist expenditures will result in an increase in real per-capita consumption for households in that development area.
- H2: The increase in real per-capita consumption is greater for households employed in tourism related industries than for those employed in other sectors.
- H3: Households in low and mid-skilled occupations will experience the largest increase in real consumption.
- H4: The benefits of increases in tourism activity will accrue primarily to those in the 1st quartile of the per-capita expenditure distribution.

As development areas are collections of economically linked communities, hypothesis one predicts that an influx of tourists to a particular development area d will result in higher real consumption levels for d 's households relative to households in areas that do not experience an increase in tourism revenues. The prediction of hypothesis two stems from the fact that an increase in tourism revenue for a development area should increase the tourism industry's demand for labor and increasing wages within the sector (Moretti 2010). We would also expect there to be a smaller increase in real-consumption for households employed in non-tourism

industries like non-tourism services and manufacturing. The size of this increase would depend on factors such as increased demand-side competition in the labor market resulting in higher wages, the level of the increase in consumption of local goods and services by households owing to greater earnings, and the strength of back-linkages between tourism and other sectors. The prediction of the third hypothesis that low to mid-skilled workers will benefit most from a tourism demand is informed by the fact that households in these occupation classes make up the largest share of tourism workers. Similarly, low and mid-skilled workers are also relatively more likely than high-skilled workers to fall within the first consumption quartile, thus hypothesis four predicts that we will see the greatest increase in real consumption among households in that segment of the consumption distribution.

The level of tourism in a given development area is not however, random, and it is therefore likely that $Tourism_{dt}$ is correlated with our error term. For example, areas that have higher levels of tourism revenue may also have qualities that attract more affluent households to live there, or that lead to better economic outcomes for locals. Perhaps development areas preferred by tourists have more attractive beaches as in [Faber and Gaubert \(2019\)](#), resulting in more affluent Jamaicans choosing to reside in those locations, and this results in amenities that in turn attract tourists. The endogeneity of $Tourism_{dt}$ motivates the use of a shift-share instrumental variable(SSIV) that will provide variation that is orthogonal to the attributes of the municipality that may influence household welfare.

In the shift share equation, $g_1, \dots, g_k$ represent shocks common to all units, while $s_{i1}, \dots, s_{ik}$ are the exposure shares that vary across units. A shift-share instrumental variable takes the form:

$$z_i = \sum_{k=1}^K \underbrace{s_{ik}}_{\text{Share}} \underbrace{g_k}_{\text{Shift}}, \quad (2)$$

with the final instrument z_i being a share-weighted average of the shifts.

This shift-share instrumental variable identification strategy is best motivated through the logic of a shift-level idealized experiment. We can consider a hypothetical experiment in which we address the unobserved variables in development areas d that are likely to bias an ordinary least squares estimation of 1. Imagine that we implement a lottery that in each year randomly selects a certain number T of tourists from specific sending regions r to receive free flights to Jamaica. We are able to dictate that only people who are invited to Jamaica via the lottery are able to visit, thus the aggregate number of tourists visiting Jamaica T , and the number from each sending region T_r is determined entirely via the lottery. We can expect that regions from which a larger number of people are selected are likely to have more visitors to Jamaica relative to regions for which there are less awardees. Visitors are then decide where on the island to stay, and the amount they spend on accommodations and other goods or services. Thus only the number of tourists from each sending region is randomly determined. Because of this random assignment of awards, the shifts from year to year in the total spending on accommodations by tourists from specific sending regions g_{rt} , are unrelated to development area d attributes such as beach quality that may influence our household outcome vari-

ables of interest y_{idt} .

Our goal is to approximate the experimental ideal explained above with existing variation in the data. I exploit two facts about tourism in Jamaica to proxy for ideal shifts and achieve identification. The first is that tourists from different regions of origin vary both cross-sectionally and over time in the areas of Jamaica they prefer to visit. The second is that tourists from different sending regions visit Jamaica in different magnitudes from year to year. Exposure shares will be the share of expenditure on accommodations in a development area that is earned from tourists visiting from specific sending regions. The shifts will be the changes in total accommodation spending by tourists from that sending region r in Jamaica overall. Jamaica's main tourism markets are the United States, Canada, and the United Kingdom, with secondary markets including the Caribbean, Continental Europe, and Latin America. For the United States and Canada, the exit surveys also provide information on the states or provinces where tourists are visiting from. I divide tourist arrivals across 7 regions given by the vector $r \in (\text{Northeast U.S., West U.S., Midwest U.S., South U.S., Canada, U.K. \& Europe, Other Countries})$.

How do we justify the argument that year-to-year variation in spending by tourists from specific sending regions is uncorrelated with the error term? First, we can note that over the period 2000 to 2021 there was substantial year-to-year variation in aggregate arrivals from these regions as well as variation in where members of these groups tended to visit in Jamaica. This variation is driven by a combination of changing local economic conditions in the origin regions, changes in access to Jamaica via available air routes, shifting preferences, and other potential factors. Some of this variation can be seen in figures 4 and 11.

Second, there was significant volatility in the global economy during the study period that was a result of geopolitical and economic events such as the September 11th terrorist attacks, the 2008 Global Financial Crisis, and the COVID-19 pandemic, among others. Decisions about whether or not to travel, and where to travel are known to be influenced by factors such as inflation and debt crises (Khalid et al. 2020), risk preferences and demographics (Karl 2018), as well as the perceived distance between to a destination (Sirakaya and Woodside 2005). These shocks produced changes in travel and vacation patterns, and these changes may have differed depending on economic and political characteristics of particular regions of origin.

Thus, we obtain year-to-year variation in the arrivals of tourists from different regions of the world. Coupled with the differential levels of exposure of Jamaican development areas, these changes in sending region arrivals generates variation in local accommodations expenditures that is orthogonal to the characteristics of those localities that may influence the outcome variables of interest.

In the tourist region of origin based instrumental variable below $(g_1, \dots, g_k)$ are shifts that are common to all units (development areas). The vector $(s_{i1}, \dots, s_{iK})$ are the exposure shares that vary across units. The term $Tourism_{dt}$ represents total tourist accommodation spending in development area d , in year t . This is equal to the sum of each region of origin r 's expenditure on accommodations in the development area in year t : $Tourism_{drt}$.

**Total Annual Accommodation Expenditures In Jamaica
and Contribution of Tourists From Different Regions 2000-2021**

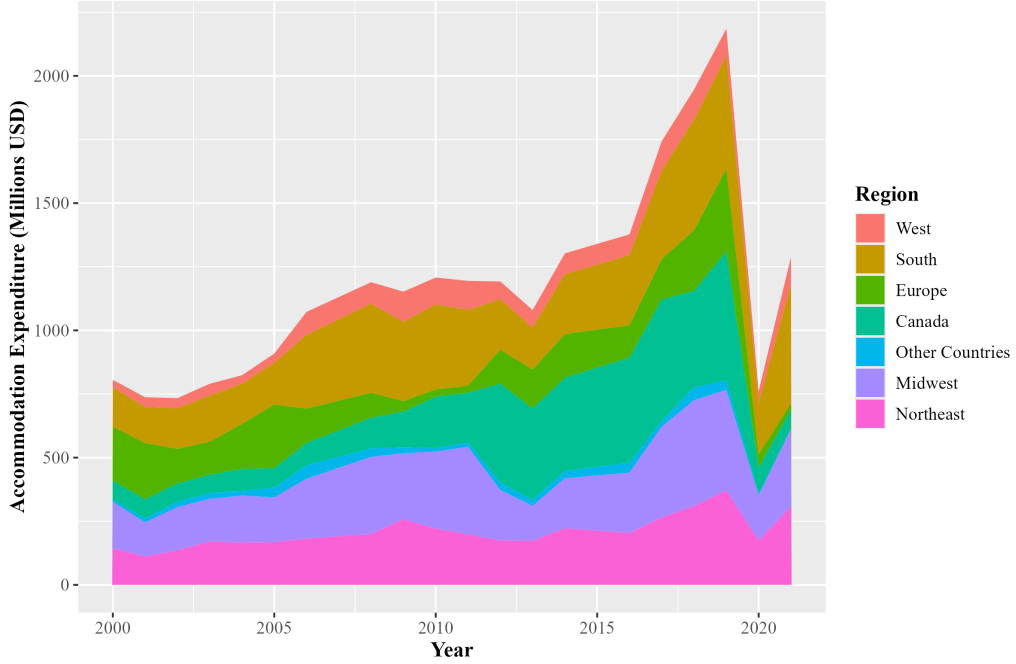


Figure 4: Level of Expenditures By Sending Region

Notes: Data comes from the Ministry of Tourism exit surveys.

$$Tourism_{dt} = \sum_{r \in R} [Tourism_{drt}] \quad (3)$$

Area d 's exposure to region r tourists in year t is given by

$$s_{drt} = \frac{Tourism_{drt}}{\sum_r Tourism_{drt}} = \frac{Tourism_{drt}}{Tourism_{dt}} \quad (4)$$

With all development areas being a part of the set $\mathbf{D}$, total spending in Jamaica on accommodations by tourists from region r in year t is therefore:

$$T_{rt} = \sum_{d \in \mathbf{D}} [Tourism_{drt}]. \quad (5)$$

We can then define the “shift”, the change in total expenditures by tourists from region r between period $t = 1$ and period $t = 0$ as:

$$g_{r1} = \frac{Tourism_{r1} - Tourism_{r0}}{Tourism_{r0}}. \quad (6)$$

The shift-share instrument therefore is given by:

$$z_{dt} = \sum_{r \in R} s_{drt} \left[\frac{Tourism_{r1} - Tourism_{r0}}{Tourism_{r0}} \right] = \sum_{r \in R} s_{drt} g_{r1}. \quad (7)$$

The shifts are constructed using the “leave-one-out” construction. In this approach, discussed and employed throughout the shift-share literature (Borusyak et al. 2022; Goldsmith-Pinkham et al. 2020; Autor et al. 2013), the shift-share instrument is written:

$$z_{dt} = \sum_{r \in R} = \sum_{r \in R} s_{drt} g_{r1, -d}, \quad (8)$$

meaning the calculated change total spending in Jamaica by the region r tourists does not include group’s change in spending for area d . This construction is meant to avoid bias in the instrument for area d potentially caused by including area d shifts. In their paper, Autor and Duggan (2003) show that including own-unit shifts could lead to a substantially stronger instrumental variable, though Goldsmith-Pinkham et al. (2020), show that the leave-out correction has a relatively minor effect when shifts average over many observations.

Over time tourists from different regions of origin have shifted the frequency with which they visit particular areas of Jamaica. For example in figures 5a and 5b we can see variation in which parts of the island Canadians preferred to visit between 2000 and 2019. I show the same figure but for tourists from the South-eastern United States in the appendix.

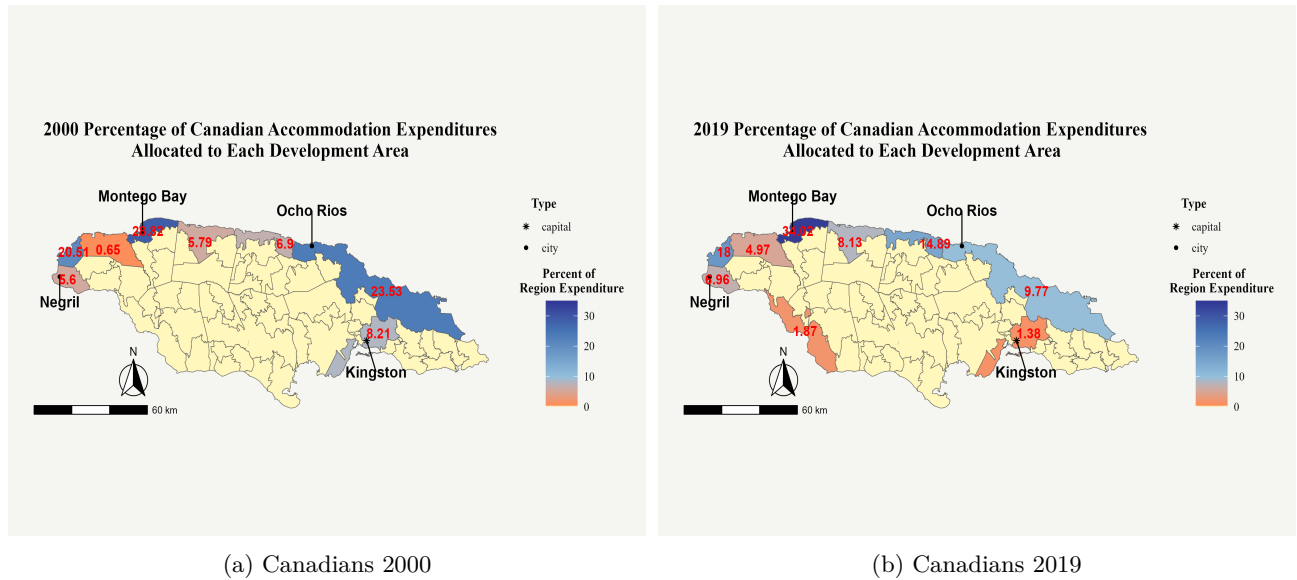


Figure 5: Variation Over 2 Decades In Spatial Distribution of Canadian Tourist Expenditures

Notes: Data comes from the Ministry of Tourism Exit Surveys.

For the 2SLS specification, Y_{idt} will be either per-capita expenditure, or an indicator for poverty status (0: Above Poverty Line, 1: Below Poverty Line) for household i , in development area d , in year t , measured in both logs and levels. Tourism intensity is given by $Tourism_{dt}$, and is calculated as total expenditures on accommodations in a development area in a given year, Z_{dt} is the shift-share instrument, η_{idt} is an idiosyncratic error term for the first stage, and second stage terms are the same as stated in the structural equation.

Stage 1:

$$Tourism_{dt} = \chi + \phi Bartik_{dt} + \iota X_{idt} + \omega D_t + \pi C_d + \eta_{idt} \quad (9)$$

Stage 2:

$$Y_{idt} = \alpha + \beta Tourism_{dt} + \psi X_{idt} + \rho D_t + \lambda C_d + \epsilon_{idt}, \quad (10)$$

Within the vector of household controls X_{idt} is the number of members in a household, the sex of the household head, and whether or not the household is located in a rural enumeration district. As the identifying variation occurs at the development area level, and because I expect the residuals of households within the same development areas to be correlated, I cluster the standard errors at this level, also.

5.1 Identification Checks

The key identifying assumption in using this shift-share instrument is that year-to-year variation in the accommodations expenditures of tourists from specific regions of origin r visiting Jamaica are uncorrelated with unobserved characteristics of the development areas or the households within them that may impact our outcome variables of interest. That is, whatever factors are driving the changes in spending by different tourist groups, and/or their decision of where they choose to spend their vacations in Jamaica, are not correlated with the unobserved characteristics of households and development areas I observe. Formally this can be represented as

$$E[Z_{dt}\epsilon_{idt}] = 0 \quad (11)$$

In order for the instrument to be valid it must satisfy relevance and the exclusion restriction. The first stage regressions for the baseline findings show that instrument functions well with a first-stage of 29 for the specification including all controls and dummy variables. This strength is also reflected in graphs of the correlations shown in the appendix.

In order to satisfy the exclusion restriction the instrument must only affect household welfare through tourist expenditures on accommodations conditional on controls. This would be violated if the instrumental variable impacts per-capita expenditures through channels such as cost of living. For example, the instrument may also be correlated with higher per-capita expenditures through its effect on the prices of locally produced services as a result of tourist spending in the community. To account for such a channel I inflate or deflate expenditures according to Jamaican regional price indices. As I also normalize all expenditures to 2024 US

dollars, the expenditure outcome variables capture real per-capita spending behavior and should not reflect price changes induced by local tourism activity.

I also use an Instrumental Variable Quantile Regressions (IVQR) to more precisely estimate effects of shocks to development areas for household's across the expenditure distribution. Given that the main dataset is a repeated cross-section, the decile within which a household falls is likely endogenous, IVQR provides a more accurate representation of the distributional impacts of the tourism shocks. I utilize the method based on [Kaplan and Sun \(2017\)](#), and perform the regressions with the associated SIVQR command in Stata ([Kaplan 2023](#)). Because this method utilizes a distance minimization technique that requires a sufficient number of observations for each quantile conditional on controls and dummy variables, I am unable to estimate the quantile regression with the full set of right hand side variables from the 2SLS specification. Instead, I employ dummy variables for each of Jamaica's 14 parishes, and a dummy variable indicating whether the observation is before or after the year 2010.

Interpreting β

It is beneficial for us to now consider the possible implications of migration on our interpretation of β . We may be concerned that β captures the effects of both existing local populations as well as Jamaicans who migrate to an area in response to the labor demand shock. We may also be concerned that the effects of tourism earnings are not captured in cases where a worker migrates from a rural area or a community with no tourism and then uses earnings from tourism to provide for family in that non-tourism producing locality. This would imply that $Tourism_{ct}$ variable for a the non-tourism producing area c is understated. Therefore, $\hat{\beta}$ would be attenuated by this under counting of development area tourism revenues. We can thus interpret the estimates of β as a lower-bound on the true effect of tourism revenues on households.

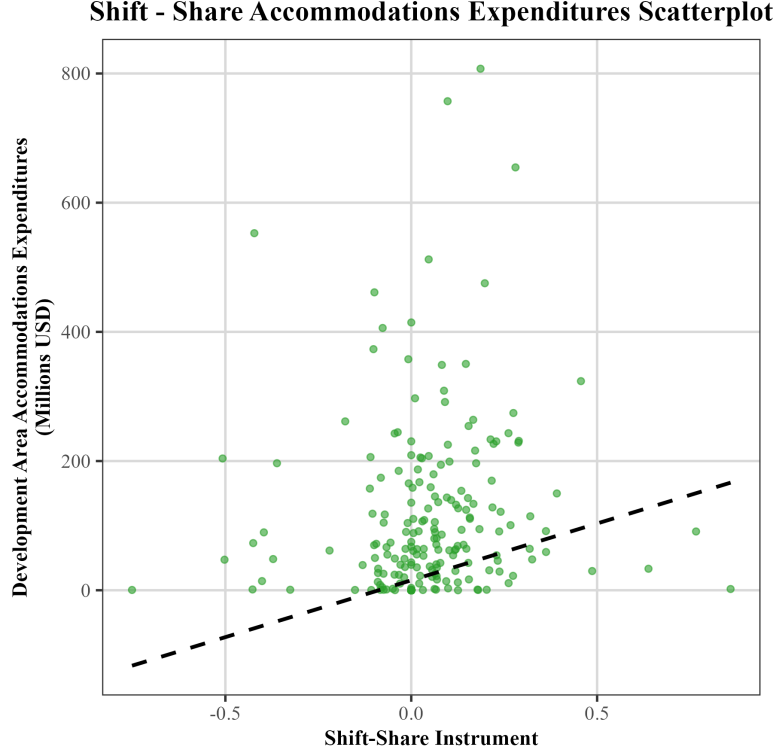
5.2 Shift-Share Instrument Diagnostics

I correct standard errors according to the methodology of [Adão et al. \(2019\)](#). They propose two approaches to calculating standard errors, and 95% confidence intervals that correct for these biases. I will refer to these calculation approaches as AKM and AKM0 following the terminology in their paper. In each of these calculations I consider shock-level variation, and cluster the shocks based on the country.

I calculate both of these standard errors for each regression in addition to the baseline clustered standard errors and bootstrapped standard error calculations. Neither of these approaches fundamentally change the findings. A sample of the all the standard errors from the baseline regressions can be seen in [table 13](#).

Another concern is that in panels or repeated cross-sections shift-share instruments there may be correlation in the shocks across periods. If this is the case, the analysis would suffer from omitted variable bias, with past shifts influencing present outcome variables. One solution they provide is to extract the idiosyncratic component of each shift before constructing the instrument ([Borusyak et al. 2024](#)). I de-mean shifts

Figure 6: Scatterplot of the Shift-Share IV and Tourist Accommodation Spending



over the study period for each region of origin r , and I then utilize the de-meanned values for each period in constructing the shift-share instrument. I provide a comparison of the two versions of the shocks in appendix 8.1.

6 Results

I now present the main findings of the analysis, guided by the four testable hypotheses drawn from the context and literature.

6.1 Baseline Findings and Welfare Outcomes

Table 3 displays findings for regressions addressing the foundational question of whether or not increases in tourism revenues produce real increases in per-capita consumption for local households. Log per-capita expenditure is the outcome variable for these regressions. The regression results are separated by whether households reside in an urban sub-district or a rural sub-district. The F-Statistic for the urban sample regression is the strongest at 70, followed by the full sample F-Statistic at 53, and the rural F-Statistic of 22. The coefficient on tourism expenditures for the urban sample 0.027, and is statistically significant at the 1% level. The log-level specification of the regression implies that an increase of ten million U.S. dollars in real development area tourism revenues increases real per-capita expenditures by 2.7% for households in the development

area relative to households in areas that do not receive this shock. There is no statistically significant effect of increased tourism earnings on rural households, as is shown in column 2. Though the first stage of the rural regression is relatively weak, the full-sample regression in column 3 lends support to the results from column 2, given the stronger full-sample first stage. The coefficient on tourism expenditures is again insignificant for the regression utilizing the entire sample.

Is the effect for urban households economically significant within the context of Jamaica? This depends in part on the typical magnitude of the variation in development area tourism expenditures. From table 21, we can see that the median change in tourism levels for a development area from one year to the next is 7.92 million U.S. dollars, and over the course of 5 years is 31.06 million U.S. dollars. Therefore, the median development area in Jamaica should see increases in tourism activity raising per-capita consumption levels by 2.1% on average. Over the course of a 5-year period, increases in local tourism intensity would produce an 8.3% increase in real per-capita expenditures on average for households in the area. In dollar terms, the average urban household's annual per-capita consumption expenditure increases by 97 U.S. dollars as a result of an increase in 10 million U.S. dollars in tourism earnings for the development area. This can be compared against the average growth of GDP per capita in Jamaica over the period of this study period of .2 percent. The first hypothesis that greater tourism earnings generate greater consumption is shown to be true, but only for urban Jamaicans.

Considering only the changes in real per-capita expenditures as the outcomes of interest may result in the analysis inaccurately capturing variation in household welfare, the main outcome of focus for this work. In order to more holistically capture household well-being and living conditions, I replicate the baseline regressions with disaggregated expenditure categories as the left-hand-side variables. The first three regressions consider food, non-food, and non-consumption expenditures. The fourth, fifth, and sixth regressions deal with three important categories of household well-being: Health, human-capital investments, and financial security. Together these regressions provide a more complete picture of how consumption is changing, and can provide additional insight on the extent to which these observed findings can be interpreted as changes in household well-being.

Table 3: IV: Relationship Between Tourism Earnings and Log Per-Capita Household Expenditure

	Annual Log Per-Capita Expenditure		
	Urban Areas (1)	Rural Areas (2)	Rural and Urban (3)
Panel A: Second Stage			
Tourism Expenditure (Tens of Millions USD)	0.027*** (0.005) [.01429, .03908]	-0.003 (0.010) [-0.024,0.017]	0.011 (0.007) [-0.003,0.026]
Observations	11447	19163	30610
Number of Clusters	34	57	59
Bootstrapped Hypothesis Test	Yes	No	No
Panel B: First Stage			
Shift-Share Instrument	11.889*** (1.418) [9.109,14.668]	11.882*** (2.544) [6.896,16.868]	11.920*** (1.633) [8.719,15.122]
First-Stage F-Statistic	70	22	53
Observations	11447	19163	30610

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 4's first and second columns show that for urban households the coefficient on tourism expenditures is positive and statistically significant for real food and non-food consumption. The coefficient in the food regression is significant at the .1% level, the coefficient for the non-food regression is significant at the 5% level. In terms of magnitude, individual food expenditures increase by 2.3% on average per 10 million dollars in area tourism revenues, while individual non-food expenditures increase by 2.9% on average. The difference between these coefficients is not statistically significant, altogether indicating that growth in local tourism revenues produces equivalent percentage increases in real food and non-food spending by locals. Though the percentages are statistically equivalent, the value of the increases in real-spending are not equivalent for each of these categories. Since the average per-capita annual food expenditure for urban households is 1613 U.S. Dollars, and is 1900 for non-food expenditure, the dollar value of the increased expenditures is 37 dollars for food, and 57 dollars for non-food goods. Approximately 61% of the increase in average consumption is spent by households on non-food items, indicating that exposed households use much of their additional consumption to spend beyond basic food items. Column 3 of table 4 shows that there is no change on average in per-capita non-consumption expenditures resulting from increased tourism activity.

Table 4: IV- Effect of Tourism On Selected Measures of Household Well-Being -Urban Households

	Log-Expenditure Outcomes			Level Welfare Outcomes		
	Log Per-Capita Food (1)	Log Per-Capita Non-Food (2)	Log Per-Capita Non-Consumption (3)	Per-Capita Healthcare (4)	Per-Capita Education (5)	Loan Repayment (6)
Tourism Expenditure (Tens of Millions USD)	0.023*** (0.003) [.01853, .02983]	0.029** (0.010) [.003635, .05586]	0.023 (0.033) [.08037, .0884]	19.770** (5.656) [4.143, 35.02]	-1.443 (4.498) [11.94, 11.29]	24.393 (9.746) [7.233, 43]
First-Stage F-Statistic	70	72	77	70	70	71
Observations	11438	11432	7217	10599	10599	6951

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

The fourth column of table 4 uses per-capita expenditure on medical services and products as the outcome variable. An increase of 10 million U.S. Dollars in tourism receipts for an area raises real expenditure on health goods and services by 19 U.S. dollars for Jamaicans in urban sub-districts. This is statistically significant at the 5% level. Columns 5 and 6 use per-capita expenditures on education goods and services, and annual household expenditures on loan repayments and interest payments, and the coefficients on tourism revenues for both are statistically insignificant. These results indicate the validity of the first testable hypotheses for urban contexts, and in particular suggest that tourism growth is improving the welfare of households in the immediate vicinity of tourist zones. Increased revenues result in higher real consumption levels for both food and non-food items, with the majority of this increased consumption going to non-food goods, and approximately one third of the increase in non-food expenditures being used by households to increase consumption of medical goods and services.

Table 5: IV: Relationship Between Household Poverty and Development Area Tourism Revenues

	Likelihood of Household Being Below the Poverty Line		
	Urban (1)	Rural (2)	Full Sample (3)
Tourism Expenditure (Tens of Millions USD)	-0.006** (0.002) [-0.010, -0.002]	0.002 (0.004) [-0.008, 0.011]	-0.002 (0.002) [-0.007, 0.003]
First-Stage F-Statistic	70	22	53
Observations	11448	19164	30612

Notes: The outcome is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45 as is the case for the urban sample. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Table 5 shows that increased tourism earnings for a development area also reduces the likelihood of a household falling below the poverty line in that area, but once again this effect is only evidence for urban dwellers. An increase in tourism earnings of 10 million U.S. dollars results in a statistically significant .6 percent lower likelihood of an observed urban household being below the poverty line relative to other localities. Taken together with the results on individual consumption categories, these results partially support the first testable hypothesis: Local urban households do derive welfare benefits from increases in their community's tourism levels, with real gains in consumption level, purchasing of health related services, and a lower likelihood of poverty. It is not clear, however, that these same benefits extend to rural households within the same development area. There are no observed changes in any categories of real consumption, or in the likelihood of poverty. At the same time, the shift-share instrumental variable is relatively weaker when evaluating the rural sample as a result of most rural households residing in areas of the country with little to no tourism. Therefore, the lack of effect of increased tourism earnings for rural households can be interpreted with some caution

6.2 Heterogeneous Impacts In The Labor Force

The next two testable hypotheses addressed questions of which workers benefit from tourism shocks. In table 6, I present results based on whether or not a household is employed in a tourism related industry. In order to confirm the predictions on employment in the literature, in the first column I regress an indicator variable that takes a value of 1 if a household works in tourism industry on development area tourism earnings. as expected, the effects are positive and statistically significant. Columns 2 and 3 repeat the baseline regression but column 3 includes only the sample working in non-tourism industries, and column 3 is only made up of the sample of tourism workers. In contrast to the predictions of the 2nd testable hypothesis, only households in non-tourism sectors experience increases in real consumption levels as a result of the increase in tourism. The coefficient of 2.9 percent implies that the baseline findings are driven entirely by tourism non-tourism workers. Similarly for poverty, columns 4 and 5 reveal that the effects of tourism on poverty only occur for non-tourism workers. Thus while employment in tourism certainly increases, actual consumption levels do not appear to be increasing. The first-stage of the tourism worker sample regression is not as strong as that of the non-tourism sector in part because of fewer observations when limiting the sample to the tourism workforce alone. Thus it may be best to interpret these findings as confirming an effect in non-tourism sectors but being unable to confirm an effect within the tourism industry rather than as a definitive indication that tourism workers are not benefiting directly.

Table 6: IV - Results By Employment in Tourism Related Industries - Urban Households

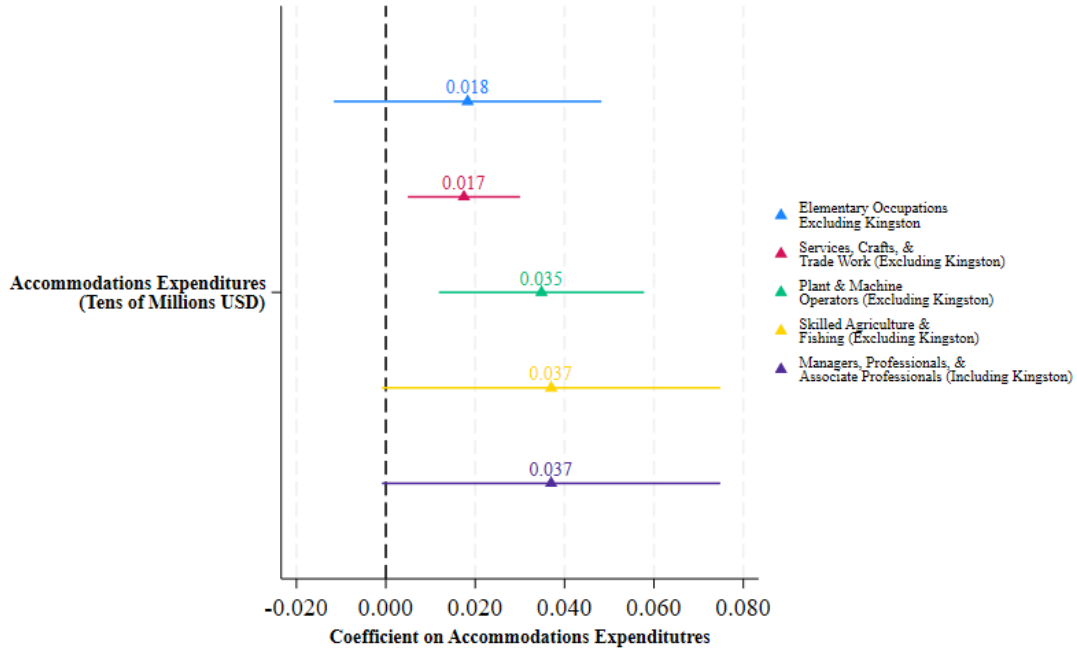
	Employment	Real Log Per-Capita Expenditure		Likelihood of Being The Below Poverty Line	
	Likelihood of Employment in Tourism (1)	Non-Tourism Related Industry (2)	Tourism Related Industry (3)	Non-Tourism Related Industry (4)	Tourism Related Industry (5)
Tourism Expenditure (Tens of Millions USD)	0.010** (0.003) [0.003,0.016]	0.029*** (0.005) [0.018,0.040]	-0.010 (0.010) [-0.030,0.010]	-0.006** (0.002) [-0.009,-0.002]	-0.006 (0.004) [-0.014,0.003]
First-Stage F-Statistic	70	69	29	69	29
Observations	11448	10390	1057	10391	1057

Notes: The Tourism Industry outcome is binary variable with 1 indicating the household head or principal earner works in a tourism related industry and 0 indicating the individual works in a non-tourism related industry. The outcome for the likelihood of being below the poverty line is a binary variable that takes a value of 1 if a household is below the poverty line and 0 otherwise. Each result above includes the coefficient estimate, followed by the standard error in parentheses, and finally the 95% confidence interval corrected with a wild bootstrap if the number of clusters is under 45. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

The third testable hypothesis predicts that because of the large share of low-skilled workers in the tourism sector, these worker will benefit the most from a positive shock to tourism in a community. Table 6 repeats the baseline regressions for separate samples of different occupations across the different skill-levels for urban households. The first column shows that for unskilled occupation there is actually no effect of tourism on annual real per-capita expenditures. These occupations require less than a secondary level of schooling. What we can see is that it is actually occupations in the level 2 skill segment that are benefiting the most, namely, “Skilled Trades” and “Plan and Machine” operators who both see positive effects of tourism. There is no statistically significant effect for skilled agricultural jobs, which is consistent with the lack of observable findings for rural households. The skill level 3 and 4 occupations which are managers, professionals and similar types of positions also shows no statistically significant effect of tourism on per-capita expenditures in line with the expectations of the third testable hypothesis.

In general these findings suggest partial support for the labor force related hypotheses. These are also consistent with the earlier observed findings. On the one hand we can see that as expected in the literature, there are statistically significant effects on the real consumption levels of households in local service sectors, and there is in fact an effect in the tradable manufacturing sector. On the other hand, the lack of an observable effect for real consumption or poverty among the tourism labor force is at odds with the original hypothesis. This finding is particularly intriguing given the prediction that the greatest benefits would be experienced by tourism workers relative to workers in other sectors of the economy. According to [Moretti \(2010\)](#), the predicted increase in earnings for workers is contingent upon imperfect geographic labor mobility and perfect mobility of labor between sectors. A lack of an effect on tourism sector workers could suggest that tourism workers are not in fact mobile between sectors and/or have a relatively high level of geographic mobility. As a result of data limitations it is not possible for me to consider these mechanisms in the context of this study. The finding that skill-level 2 occupations are the ones that see the greatest increase in their consumption levels from an increase in tourism would, however, be consistent with a high level of sectoral mobility. This finding partially supports the third hypothesis regarding the occupations that would benefit from

Figure 7: Cross-Section Comparison of IV Coefficients Across Occupation Categories In Urban Areas



tourism, as unskilled occupations are shown to experience no increase in real consumption levels.

6.3 Heterogeneous Effects Across The Consumption Distribution

Finally, I present results on the question of where on the consumption distribution the benefits of tourism revenues accrue. The fourth testable hypothesis predicted that this effect would also operate through the skills distribution of the labor force. Because households working in tourism, and working in skill-level 1 and 2 occupations are disproportionately likely to be in the 1st quartile of the Jamaican expenditure distribution, greater tourism revenues are predicted to produce the greatest impacts on Jamaicans in the 1st quartile. considering the fact that the previous results could not confirm an effect on tourism sector workers, the accuracy of the fourth hypothesis is not necessarily as certain.

I choose to test the fourth hypothesis using an Instrumental Variable Quantile Regression (IVQR). A quantile regression approach allows for a more reliable analysis differences in effects based on consumption than say, separating the sample by quartile and then conducting individual regressions for each quartile. The IVQR analysis for this paper follows the methodology of [Kaplan and Sun \(2017\)](#), using the associated SIVQR STATA package ([Kaplan 2023](#)). I once again use the urban sub-sample of the main repeated cross-section household dataset. IVQR uses a minimum distance procedure that is reliant on there being a sufficient number of observations in each quantile conditional on controls and dummy variables. The urban sample is not large enough to run the IV Quantile Regression approach when using the full set of baseline controls and dummy variables. As such, for the IV Quantile Regression analysis I limit the controls to the household size,

the sex of the household head, and a binary variable indicating whether or not the year of the observation is less than or equal to 2010.

Figure ?? presents the results for the analysis. It shows that among the urban sample the lowest quartile of the expenditure distribution is where all of the statistically significant effects of increases in tourism are concentrated. This aligns with the fourth hypothesis, and aligns with the previous finding that positive tourism shocks reduced the likelihood of households being below the poverty line. That being said, it is important to consider where on the national expenditure distribution those in the lowest urban quartile fall in order to full evaluate which segments of households are benefiting from tourism. These findings indicate that though the urban poor are benefiting most significantly, when we consider the effects in context, they are still mostly accruing to those in the middle of the expenditure distribution. Thus when evaluated against the goals of the tourism sector this implies that we can confirm impacts only on the middle and lower-middle segments of the national expenditure distribution. This aligns well with the goal of improvements in tourism generating improvements in a broad cross-section of the population, but also implies that the poorest households may not be experiencing economically or statistically significant benefits in the short-run from tourism booms.

Table 7: IV: Quantile Regression of Log Per-Capita Consumption Expenditure on Tourist Accommodations Expenditures - Urban Households

	Per-Capita Expenditure Quantiles		
	25 (1)	50 (2)	75 (3)
Tourism Expenditure (Tens of Millions USD)	0.004*** (0.001) [0.002,0.006]	0.001 (0.001) [-0.001,0.004]	-0.001 (0.408) [-0.800,0.799]
Observations	11125	11125	11125
Smoothing Bandwidth	95	95	95

Notes: The above columns represent the coefficient estimates for the 25th, 50th, and 75th quantiles of the Log per-capita expenditure quantiles. Each regression includes a dummy variable indicating whether or not the year is less than or equal to 2010 or not.

6.4 Robustness To Alternative Explanations

I now aim to provide plausible explanations for the mechanisms underlying these observed effects. In order to do this I employ the panel dataset. This does not allow for a one-to-one comparison of the findings because of limited observations, but it still can provide useful insight on the labor force and how it is affected by tourism booms. The panel regression is useful because it allows me to limit my regressions to household who are located in the same industry for each period in which they are observed. In contrast, the cross-sectional results in table 6 reflect the effects of tourism shocks on incumbent workers in an industry, as well as those who are newly hired. The panel regression also allows for comparison of the effects of tourism on households depending on the segment of the distribution in which they fall in the first period they are observed. This can provide a more precise view of where on the expenditure distribution households are experiencing the greatest improvements in consumption because of a tourism shock. Unfortunately, because the unbalanced panel has

fewer observations than the cross-section in order to maintain statistical power in the first and second stages, I keep both urban and rural communities households as well as Kingston in this sample. Therefore, this regression does not provide a one-to-one comparison with the main results of the paper, though I argue that its results can still be informative.

Table 8 displays the results of the fixed effects instrumental variable regression of annual real per-capita expenditures on development area tourism revenues, with disaggregations according to selected sectors and the quartiles in which households fall. The first column shows results for the regression using the full sample with all sectors and all quintiles. As expected, on average there is no effect for tourism revenues on household expenditures. Columns 2-4 show results for households that work in either “Non-Tourism Services” or “Manufacturing” during all periods they are observed. Column 2 and column 4 present findings for the 1st and 4th quartiles, respectively, but both have weak first stages and it is therefore not possible to draw strong conclusions from them. Column 3 has a strong first stage and is the subset of households that who fall in the 2nd and 3rd expenditure quartiles during the first period that they are observed. For these households the coefficient on tourism earnings is 2.5 percent per 10 million dollar increase in revenues. This finding is significant at 0.001 level.

The final three columns are regressions for households that work in tourism for at *at least* one of the periods in which they are observed. Once again, the regressions for the 1st and 4th quartiles in the 5th and 7th columns present with weak first stage results making serious interpretation difficult. Column 6 presents findings for households in either the 2nd or 3rd expenditure quartiles in their first period, and the coefficient is again positive and statistically significant. The magnitude is 2 percent. These findings shed further light on the industries in which workers benefit the most from local tourism shocks. It further supports the quantile regression results by indicating that it is Jamaican households from the lower middle to the upper middle of the national consumption distribution that are experiencing the greatest benefits from tourism booms. The 6th columns leaves open the possibility that there is in fact some impact for tourism workers in the industry, showing that a household with at least one period in the tourism industry does indeed benefit from a tourism shock. There are too few tourism employed households to limit the sample to only that segment of the workforce, so it is not possible to precisely determine the effects for households only employed in tourism across their observations.

Table 8: IV-Panel Per-Capita Consumption by Industry

	All Sectors	Non-Tourism Services & Manufacturing			Ever Employed in Tourism		
	All Quartiles	Quartile: 1	Quartiles: 2-3	Quartile: 4	Quartile: 1	Quartiles: 2-3	Quartile: 4
DA_EXP_Scaled	-0.003 (0.007) [-0.017,0.011]	0.083** (0.038) [0.006,0.159]	0.025*** (0.008) [0.008,0.042]	-0.030* (0.015) [-0.061,0.000]	0.091** (0.041) [0.008,0.175]	0.020** (0.009) [0.002,0.038]	-0.022 (0.018) [-0.058,0.013]
First-Stage F-Statistic	26	8	64	13	11	59	13
Observations	6032	313	1133	713	225	826	543
Number of Clusters	58	45	53	45	39	46	38

Notes: Non-Tourism Services and Manufacturing observations are those households that are employed in either of those sectors for all periods they are observed. Ever Employed in Tourism is a sample of households that are employed in a tourism related industry for at least one of the periods in which they are observed. Each result above includes the coefficient estimate, followed by the standard error in parentheses. Regressions are separated by the initial quartile in which a household is observed. All regressions include a vector of household fixed effects and year fixed effects. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

Next we can consider the concern that the observed results are driven by the inflationary effects of the tourism shock, even when using regional price index adjusted expenditure and observing decreases in the likelihood of a household being in poverty. One prediction from the literature that seemed unlikely to be true in the short-to-medium run in the Jamaican case was the expectation of the localized shock increasing rent on average. In order to formally test for inflationary effects on rent, I regress real per-capita average monthly rent payments on development area tourism revenues for the sample of urban households. This regressions uses the main cross-section dataset. I find no effect of tourism earnings on real per-capita rent expenditures for households, as shown in table 9. The strong first Stage F-Statistic of 72 gives further support to the accuracy of the null effect. To further confirm that the main results are not merely capturing inflationary pressures, I run regressions disaggregated by the employment type of a household. This refers to whether the household works in the private sector, for the government, or as an own-account worker. While the wages of private sector and own-account employees are likely to be affected by short-run changes in market forces in their local communities, government employee salaries are determined by the national budget approved by the Jamaican Parliament for each fiscal year. Therefore, short to medium run inflationary effects can be expected to result in decreases in real per-capita expenditures for government employees whose wages will not respond to a sudden tourism shock.

Table 9: IV: Regression of Real Per-Capita Rent Expenditure on Tourism Revenues

	Real Per-Capita Rent (1)
Tourism Expenditure (Tens of Millions USD)	10.620 (6.772) [-3.158,24.397]
First-Stage F-Statistic	72
Observations	11442

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

In table 10 I show the results of expenditure regressions disaggregated by the employment type of a household for those living in urban sub-districts. Column 3 shows a lack of any effect of increased tourism earnings on real per-capita expenditures for government workers. Only those working in the private sector are shown to have an increase in real consumption levels as a result of a positive tourism shock. These results lend further support to the prediction of few inflationary effects of tourism in the Jamaican context on local communities. A lack of significant negative consumption impacts for non private sector employees is promising as it suggests a lack of the crowding out effects observed by [Allen et al. \(2021\)](#) in the Barcelona setting. While this study cannot speak to long-term welfare impacts of tourism on Jamaican local communities, this does suggest that at least in the short to medium run, there are not significant welfare losses for segments of the population not involved in the tourism sector.

Table 10: IV: Real Per-Capita Consumption By Employment Type

	Real Log Per-Capita Expenditure		
	Private Sector (1)	Government (2)	Own Account Worker (3)
Tourism Expenditure (Millions USD)	0.002* (0.001) [0.000,0.003]	-0.002 (0.001) [-0.005,0.000]	0.000 (0.001) [-0.002,0.003]
First-Stage F-Statistic	59	23	32
Observations	10295	2032	11819

Notes: Each result above includes the coefficient estimate, followed by the standard error in parentheses. All regressions include a vector of household controls, development area dummies and year dummies. Tourist Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shock components are demeaned to extract the idiosyncratic component of the shocks.

7 Discussion and Conclusion

This paper investigates the effects of growth in an areas tourism revenues on the consumption and welfare of households in a developing country. The two principal questions guiding the analysis are, do increases in tourist expenditures produce increase in real per-capita consumption for local households and, to what extent are consumption increases observed among households at or below the poverty line? I combine novel micro-datasets of tourist activity and household expenditure surveys in Jamaica spanning two decades, and spatially link the observations between 2001 and 2021 across consistent spatial units, to produce a comprehensive view of variations in tourism activity and household spending across the study period. This paper employs a shift-share instrumental variable identification strategy that harnesses two key aspects of tourist behavior: Variation in the exposure of different areas of Jamaica to tourists from different regions of origin, and plausibly exogenous variation in the total spending from tourists of different origin groups in Jamaica from year to year. Armed with this identifying variation this study then analyzes the effects of variations in annual development area tourism revenues on per-capita household expenditures, poverty, and 6 disaggregated spending categories.

I find strong evidence that positive tourism revenues generate a positive and economically significant effect on real consumption levels for urban residents of the surrounding development areas. These findings are driven by households working in non-tourism services and manufacturing. Among urban households in Jamaica, increases in tourism earnings generate statistically equivalent percentage increases in per-capita food and non-food expenditures though the dollar value of increases in non-food consumption are larger than those for food consumption. These findings also indicate that households use a sizable share of their increased consumption ability to invest in a key measure of welfare, health and medical services. Combined with meaningful decreases in the likelihood of households falling below the poverty line as a result of increased tourism, I

interpret these results as indicating that tourism revenues generate improvements in the welfare of households in areas experiencing tourism growth.

The results from this study provide 3 key insights on the effects of tourism on the Jamaican labor force. The first is that as would be predicted by the literature on labor demand shocks for labor intensive industries, increases in tourism revenues generate increases in the likelihood of households being employed in the tourism industry within a given area. The second result is that the benefits of the tourism shock produce increases in real consumption for households working in non-tourism industries. There is a similar story for poverty, with there being no observable effect on households employed within the tourism sector. This result may be in part because of a lack of identifying variation in the first stage when limiting the sample to only tourism employees, but it is still notable considering long-running complaints within the tourism sector in Jamaica about low-wages and poor working conditions. I provide additional support to these findings by employing an unbalanced panel dataset of households that is a subset of the repeated cross-section I utilize for the main analysis. This regression, though not a one-to-one match with the original reveals importantly that the positive effects of tourism on consumption are felt principally by workers who are employed in either non-tourism services or manufacturing throughout each period they are observed. The panel does, however, reveal statistically significant effects for households that are employed within the tourism industry for at least one period, though lack of statistical power prevents limiting the sample to households exclusively employed in tourism across each observation.

The third labor related result presented by this paper is that the skill-level of occupations, and the specific type of occupation are particularly relevant for where in the labor force benefits of tourism are observed. Increases in an areas tourism levels exclusively benefit residents working in either “Service, Crafts, and Trades” or “Plant and Machine Operators”, both of which are skill-level 2 occupations as defined by the ISOC. This result also is consistent with the result that workers in non-tourism services and manufacturing experience the greatest benefit from tourism growth. Particularly important to the aims of the Jamaican government, and more generally to tourism based development, there are no observed effects for skill-level 1, unskilled occupations, though they represent a relatively larger share of the tourism labor force. There are also no observed effects on skilled agricultural workers, despite the aim of improved linkages to the tourism and agricultural sectors. Viewed against the goals of the Jamaican government and sustainable tourism development, tourism’s ability to drive improvements in well-being among those in mid-skilled occupations who collectively represent roughly 75 percent of employment is a positive and nontrivial finding. At the same time, the lack of effect for those who are unskilled, or perform skilled agricultural work, also demonstrates the potential limits of tourism’s inclusiveness.

Finally, in response to the second research question, this paper shows that tourism can produce improvements in consumption and well-being for households at or near the poverty line. This result is however tempered by the fact that these effects are only observed for urban households. Results of the quantile regression show that for urban communities, the positive effects of tourism on earnings are found in the 1st expendi-

ture quartile. However the real expenditure levels of urban households in the first quartile in fact span the 1st and 2nd quartiles when considering the full national sample. This is further supported by the panel industry regressions which show households in the 2nd and 3rd quartiles benefiting from increased tourism revenues. Once again this result shows both the inclusive potential for tourism services, as well as their limits. Though a broad range of urban households are shown to benefit, the lack of observed effects for rural sub-districts (where the poorest Jamaicans live), shows the difficulty of harnessing back-linkages to generate prosperity across industries and geography.

What are the implications of these outcomes for policy and research into tourism-based development? In terms of policy this paper shows that tourism can be an effective means for generating employment in the context of local communities in a middle-income country such as Jamaica. This suggests policies meant to encourage tourism investment such as those in Jamaica such as relief from taxes and tariffs on productive inputs have the potential to be welfare improving for communities directly exposed to tourism. However the extent of these policies must also be weighed against the magnitude of the economic benefits to households, and importantly, the magnitude of the change in tourism revenues needed to generate real gains in well-being. The scale of the Jamaican tourism industry assures that for the communities in which tourism takes place, the necessary increases in tourism needed to benefit households does in fact occur.

Another policy implication is the potential necessity of upskilling elements of a workforce in order to benefit from the spillover effect of tourism services. Results demonstrate that skill-level 2 occupations are consistently the beneficiaries of gains in tourism for residents of an area. Thus training programs prepare a workforce for the types of occupations that benefit from tourism demand shocks may be advisable as a way of improving the likelihood that households are capable of reaping the benefits of growth in the tourism sector.

Finally the study also shows the potential benefits from a more segmented style of tourism in a developing country context as opposed to the integrated urban style tourism seen in many industrialized cities. While there is some evidence for slight inflationary impacts of tourism among those working public sector jobs, there broadly do not appear to be significant impacts on cost of living for tourism exposed development area residents. While the style of tourism practiced in many developing contexts has often been criticized for its “enclave” like nature, in a case where tourists are likely to have significantly higher levels of income than an areas residents, this study suggests a potential positive effect may be a lack of major inflationary pressures on locals. A complete treatment of the impacts of the *type* and *structure* of the tourism industry within in a country is beyond the scope of this paper, but these findings do reinforce the potential tradeoffs policy makers must consider when deciding on tourism investments.

There are two limitations of this paper that can motivate additional research in the area of tourism services and its impacts on households welfare and local economic development. The first is an inability to precisely account for migration due to data availability. While it is broadly known via census data and general conversations with Jamaican officials that migration to the the tourism zones from rural areas is a common feature of Jamaican demographic developments, I am unable to observe this in my household datasets. Fully

quantifying the impacts of tourism on economic development will require considering the movement of individuals and/or households across regions of Jamaica for work. However, because of the presence of migration, I would consider the effects observed by this study to be a lower-bound on the effects of tourism as they demonstrate the ability of tourism to raise earnings both for new residents and long-time ones within an area.

The second limitation is that I am unable to disentangle whether the observed spillover effects of tourism to other sectors are a result of labor-demand induced increases in earnings, back-linkages from tourism increasing demand for local services and manufacturing, or the effects of an increase in an area's total wage bill resulting in increased expenditures from households or some combination of all three. Further research considering these short-to-medium run channels could be useful for testing additional policy scenarios and help to better capture which residents benefit and to what degree. This would also be useful for understanding the lack of observed impacts on households working in tourism related industries and rural households.

7.1 Conclusion

This paper has contributed principally by improving understanding of the magnitude and scope of the effects of tourism services on household consumption and well-being in a developing country. This paper has critically demonstrated that fully characterizing the impact of tourism on local economic development requires considering heterogeneity of households in terms of skill-levels and occupations, and that these in turn impact which segments of the consumption distribution benefit from positive tourism shocks. In addition, this study provides a balanced analysis of the potential for tourism to generate economically meaningful gains for major segments of the population, while also qualifying these impacts by showing limits in effects across space, and skill-level.

Tourism services has grown into a major global sector, and presents opportunities for emerging economies to exploit natural or cultural endowments to the benefit of their populations. Tourism services show promise in some instances, but in many cases have also been criticized for what is considered their exploitation of local populations. This paper has shown that real economic development gains are possible from significant segments of a local population in the presence of positive tourism shocks, while also showing that tourism based development involves important trade-offs and limitations that must be considered in order to best harness its potential for meaningful local prosperity.

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8 Appendix

8.1 Appendix 1: Instrument Summary Statistics and First Stage Regressions

One concern when using shift-share instrumental variables in a panel setting is the possibility of serial correlation in the shifts. As explained by [Borusyak et al. \(2024\)](#), if shocks g_{kt} are correlated with shocks from period $t - 1$, then the estimated effect of tourism on the outcome variables reflects both the contemporaneous effect and the dynamic effect of the previous shock.

Table 11: Summary of Exposure and Shock Measures

	Variable	Mean
1	Mean Exposure Share	0.02
2	Median Exposure Share	0.00
3	Standard Deviation Exposure Share	0.07
4	Mean Shock	0.11
5	Median Shock	0.03
6	Standard Deviation of Shock	0.34
7	Inverse HHI: Shocks	280.19

Notes: These summary statistics cover the SSIV shocks and exposure shares calculated from the Ministry of Tourism Exit Surveys.

8.2 Appendix 2: Shift-Share Instrument Diagnostics and Summary Statistics

Table 12: IV: Shift-Share Instrumental Variable Full Summary Statistics

	Year	Mean Exposure	Median Exposure	SD Exposure	Mean Shock	Median Shock	SD Shock	Inverse HHI: Shocks
1	2001	0.02	0.00	0.08	0.05	-0.01	0.29	236.76
2	2002	0.02	0.00	0.08	0.10	0.15	0.26	308.57
3	2003	0.02	0.00	0.08	0.07	0.04	0.11	280.96
4	2004	0.02	0.00	0.07	0.02	0.00	0.20	298.28
5	2005	0.02	0.00	0.08	0.16	-0.02	0.32	267.17
6	2006	0.02	0.00	0.08	0.60	0.25	0.71	267.17
7	2008	0.02	0.00	0.07	0.04	0.10	0.25	270.23
8	2009	0.02	0.00	0.07	-0.04	-0.12	0.33	264.21
9	2010	0.02	0.00	0.07	-0.04	-0.09	0.26	258.78
10	2011	0.02	0.00	0.07	0.02	-0.01	0.11	258.27
11	2012	0.02	0.00	0.07	0.60	-0.13	1.39	250.70
12	2013	0.02	0.00	0.06	-0.10	-0.07	0.14	274.58
13	2014	0.02	0.00	0.07	0.24	0.24	0.14	243.03
14	2016	0.02	0.00	0.07	0.09	0.12	0.22	252.44
15	2017	0.02	0.00	0.06	0.21	0.26	0.29	351.61
16	2018	0.02	0.00	0.06	0.32	0.18	0.44	350.13
17	2019	0.02	0.00	0.06	0.07	0.01	0.21	293.13
18	2021	0.02	0.00	0.07	-0.43	-0.26	0.41	317.39

Notes: Above are summary statistics for the SSIV instruments for all years of the analysis.

I report the AKM standard errors of a shock-level regression following the methodology of [Adão et al. \(2019\)](#) in table 13. In panels A, B, and C, the results of my baseline regression specifications are compared for all Jamaican households, urban households, and rural households, respectively. I show the implied estimates for

five different standard error specifications: Homoskedastic, White standard errors, regular clustered standard errors, and finally, the AKM and AKM0 specifications. The AKM approach corrects for potential correlation in shocks between units that have similar exposure shares. For both the AKM and AKM0 standard errors, the results of my baseline regressions remain.

Table 13: Comparison of AKM Regression Results Across Methodologies

Method	Estimate	Std.Error	P.Value	Left.CI	Right.CI
Panel A: IV Estimates All Regions					
Panel A: IV Estimates All Regions					
Homoscedastic	2e-04	5e-04	0.7102	-7e-04	0.0011
EHW	2e-04	5e-04	0.7166	-7e-04	0.0011
Reg. Cluster	2e-04	7e-04	0.8078	-0.0012	0.0015
AKM	2e-04	2e-04	0.3069	-2e-04	5e-04
AKM0	2e-04	2e-04	0.2502	-1e-04	8e-04
Panel B: IV Estimates Urban Households					
Panel B: IV Estimates Urban Households					
Homoscedastic	0.002	6e-04	0.0011	8e-04	0.0032
EHW	0.002	6e-04	0.0016	8e-04	0.0033
Reg. Cluster	0.002	4e-04	0	0.0011	0.0029
AKM	0.002***	5e-04	1e-04	0.001	0.003
AKM0	0.002***	7e-04	0	0.0013	0.004
Panel C: IV Estimates Rural Households					
Panel C: IV Estimates Rural Households					
Homoscedastic	-3e-04	6e-04	0.6459	-0.0016	0.001
EHW	-3e-04	6e-04	0.6465	-0.0016	0.001
Reg. Cluster	-3e-04	0.0011	0.7841	-0.0024	0.0018
AKM	-3e-04	3e-04	0.255	-8e-04	2e-04
AKM0	-3e-04	3e-04	0.2225	-9e-04	2e-04

Notes: The tourism expenditure is measured in millions and is measured at the level of the development area.

Next I show graphs before and after my de-meaning of the shift-share tourism arrival shocks.

8.3 Appendix 3: Panel Regression Results

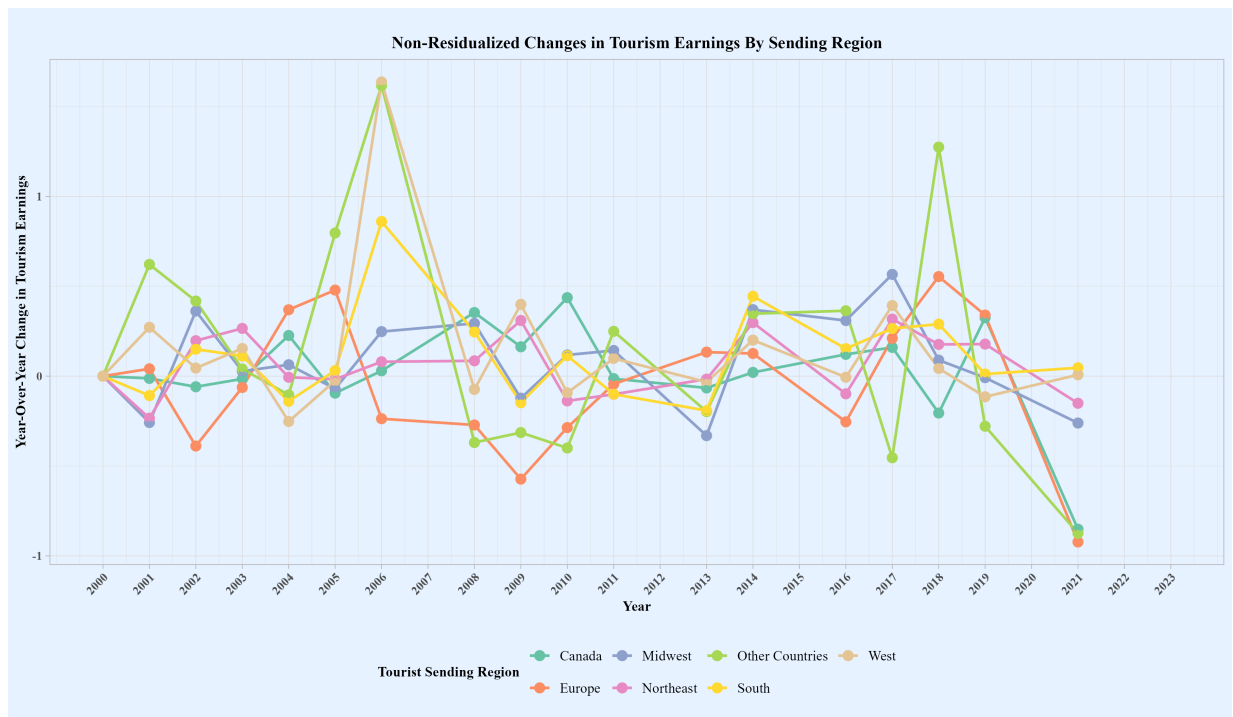


Figure 8: Graphs of Original Shift-Share Shocks

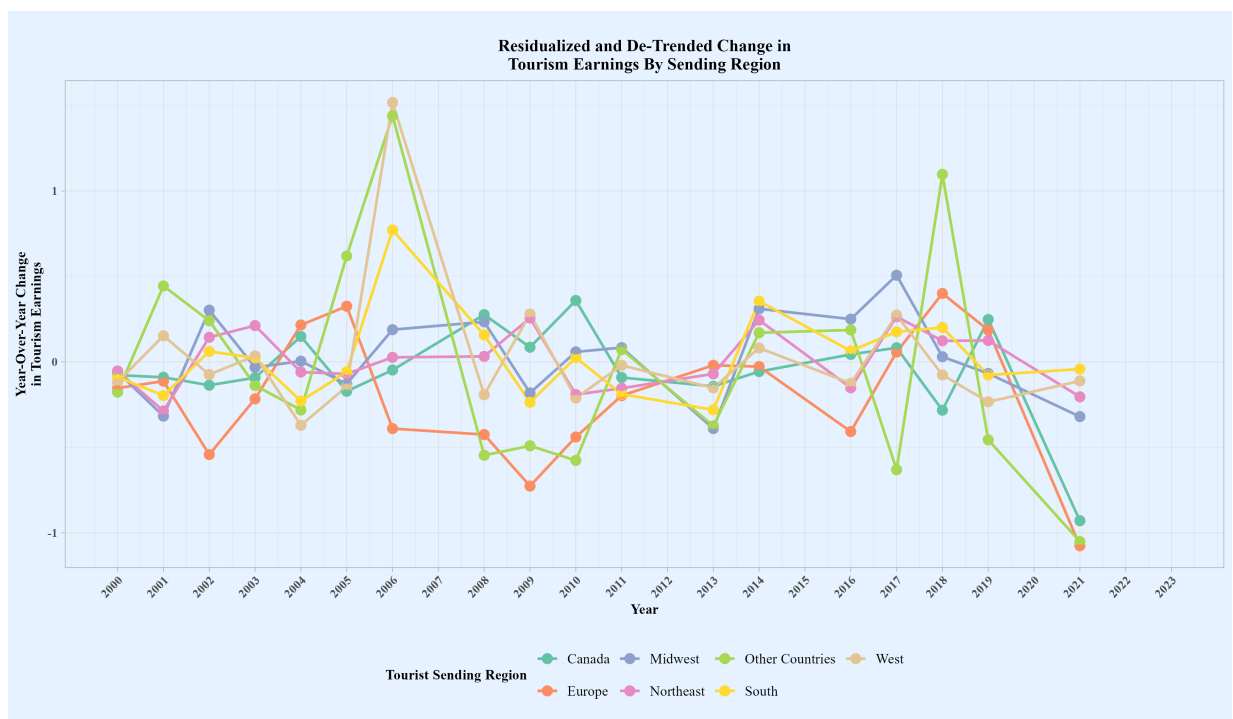


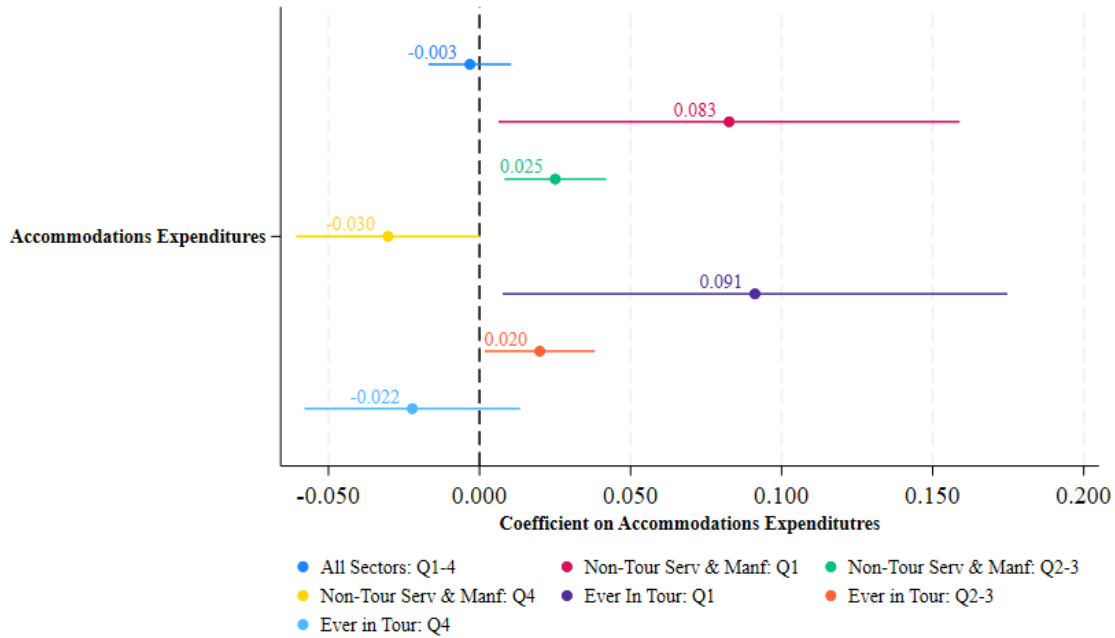
Figure 9: Graph of Residualized Shift-Share Shocks

Table 14: IV: Panel Regression of Baseline Results

	Log Per-Capita Expenditure(USD)		
	(1)	(2)	(3)
Accommodations Expenditures(Millions USD)	-2.8e-04 (7.1e-04)	-2.8e-04 (7.1e-04)	-3.2e-04 (6.8e-04)
Household Size			-2.1e-01*** (1.4e-02)
First-Stage F-Statistic	26	26	26
Observations	6032	6032	6032
Standard Deviation	0.711	0.711	0.711
Household Control	No	No	Yes
Household Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	No	Yes	Yes

Notes: Accommodation Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks. Female Household Head indicates either a single adult female or household with multiple persons for which the household head or principal earner is female.

Figure 10: Panel Comparison of Coefficients Across Industries of Employment



8.4 Appendix 4: Additional Regression Results

Table 15: IV: Non-Consumption Expenditure

	Urban Log Per-Capita Non-Consumption Expenditure	Rural Log Per-Capita Non-Consumption Expenditure
Tourism Expenditure (Millions USD)	3.2e-03 (5.0e-03) [-0.007,0.013]	3.8e-03 (2.1e-03) [-0.000,0.008]
First-Stage F-Statistic	68	10
Observations	7282	11832
Standard Deviation	1.979	1.961
Number of Clusters	39	64
HH Controls	Yes	Yes
DA Dummy	Yes	Yes
Year Dummies	Yes	Yes

Notes: Tourism expenditure is measured in millions and is measured at the level of the development area.

Table 16: IV-Relationship Between Tourism Earnings and Log Household Expenditure By Per-Capita Expenditure Decile Urban Households

	Log Per Capita Expenditure(USD) Separated by Deciles					
	(1-10)	(2-10)	(4-10)	(6-10)	(8-10)	(9-10)
Tourism Expenditure (Millions USD)	1.7e-03** (5.1e-04)	1.5e-03* (5.7e-04)	1.5e-03** (5.4e-04)	1.2e-03* (5.6e-04)	1.6e-03* (6.2e-04)	2.2e-03** (6.7e-04)
First-Stage F-Statistic	70	69	54	61	66	48
Observations	11526	11005	9747	8050	5860	4457
Standard Deviation	0.721	0.654	0.589	0.544	0.522	0.535
HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
DA Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Tourism expenditure is measured in millions. All household controls are included.

Table 17: IV - Relationship Between Tourism Earnings and Log Per-Capita Household Expenditure by Rurality

	Urban	Rural
Tourism Expenditure (Millions USD)	4.3e-04 (8.8e-04) [-0.001,0.002]	-8.2e-04 (1.2e-03) [-0.003,0.002]
First-Stage F-Statistic	61	18
Observations	17792	18381

Notes: Accommodations Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks. Female Household Head indicates either a single adult female or household with multiple persons for which the household head or principal earner is female.

Table 18: IV: Regression of Log Per-Capita Expenditure By Gender of Individual or Household Head

	Urban		Rural	
	Male-Headed Households	Female Headed Households	Male-Headed Households	Female Headed Households
Tourism Expenditure (Millions USD)	-4.6e-04 (1.2e-03) [-0.003,0.002]	1.2e-03 (1.7e-03) [-0.002,0.005]	4.8e-04 (1.4e-03) [-0.002,0.003]	1.4e-04 (1.5e-03) [-0.003,0.003]
First-Stage F-Statistic	35	20	11	9
Observations	10989	9324	13351	7716
Standard Deviation	0.723	0.725	0.722	0.682
Number of Clusters	45	45	70	70
HH Controls	Yes	Yes	Yes	Yes
DA Dummy	Yes	Yes	Yes	Yes
Year Dummies	Yes	Yes	Yes	Yes

Notes: The tourism expenditure is measured in millions and is measured at the level of the development area.

Table 19: IV- Comparing The Impact of Tourism on Urban Homeowners vs. Urban Renters (Excluding Kingston)

	Homeowner Expenditure	Renter Expenditure
Tourism Expenditure(Millions USD)	3.8e-03* (1.4e-03) [0.001,0.007]	1.1e-04 (8.1e-04) [-0.002,0.002]
First-Stage F-Statistic	13	15
Observations	6063	9613
Standard Deviation	0.706	0.743
Number of Clusters	34	60
Bootstrapped Standard Errors	Yes	Yes
HH Controls	Yes	Yes
DA Dummy	Yes	Yes
Year Dummies	Yes	Yes

Notes: Bootstrapped standard errors are provided in parentheses below the coefficient estimates. The bootstrapped confidence intervals are in brackets below the standard error estimates. Accommodations Expenditures are calculated at the development area level in tens of millions of 2024 U.S. Dollars. Household expenditures are inflated or deflated based on Jamaican regional price indexes to obtain real consumption levels across different parts of the country. All shift-share instrument shocks are demeaned to extract the idiosyncratic component of the shocks.

8.5 Appendix 5: Tourism Summary Statistics

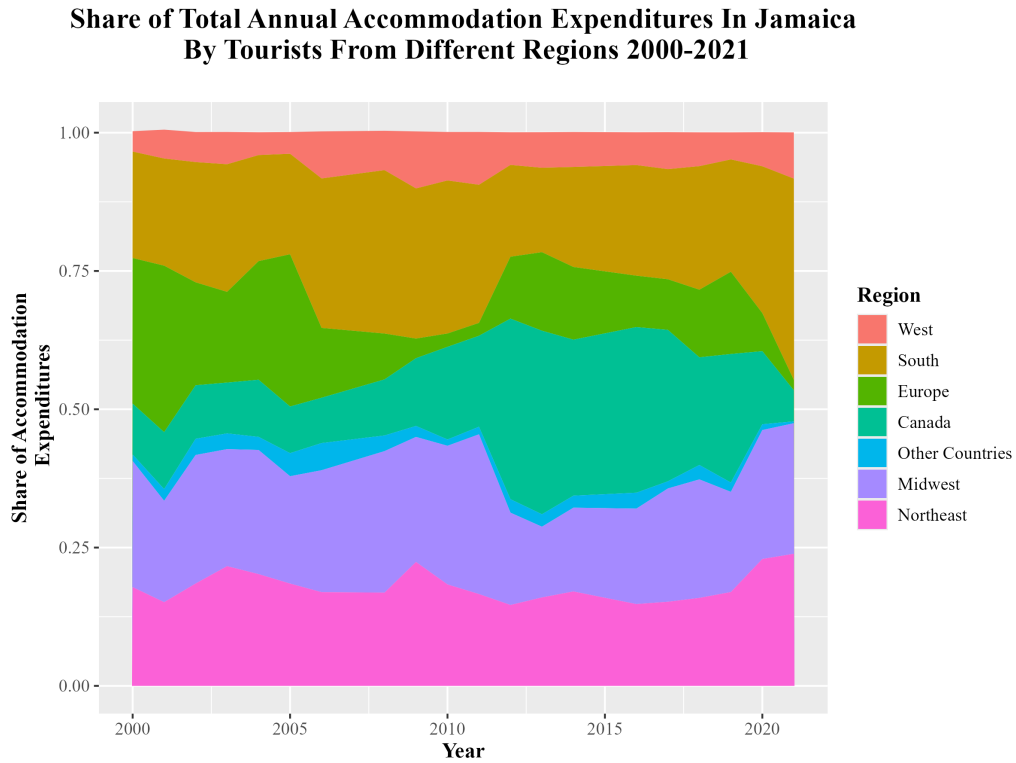


Figure 11: Share of Expenditures by Region

Table 20: Tourist Summary Statistics

	Mean	Standard Deviation	Median
Avg. Accom Price per Person	590.03	495.58	520.69
Number of People in Party	1.95	0.96	2.00
Total Cost of Trip	2706.91	1845.23	2350.90
Length of Stay	7.47	5.88	7.00
Visit for Vacation	0.74	0.44	1.00
Return Visitor	0.47	0.50	0.00
Summer Visitor	0.57	0.49	1.00
Income Over US60,000	0.50	0.50	0.00
Observations	78774		

Source: Author's own calculations based on Ministry of Tourism Exit Surveys (2000-2023).

Table 21: Variation In Tourist Expenditure Levels Over Time Lags

Statistic	1 Year Lagged Value	5 Year Lagged Value	10 Year Lagged Value
Mean Expenditure Change	21.09	46.04	63.53
Median Expenditure Change	7.92	31.06	36.73
Min Expenditure Change	0	0.02	0.02
Max Expenditure Change	235.54	430.92	449.26
SD Expenditure Change	33.35	63.34	89.45

Notes: All expenditures are in millions of US Dollars corrected for inflation to the year 2024.

8.6 Appendix 6: Additional Maps And Geographic Summary Statistics



Figure 12: Map of Jamaican Parishes

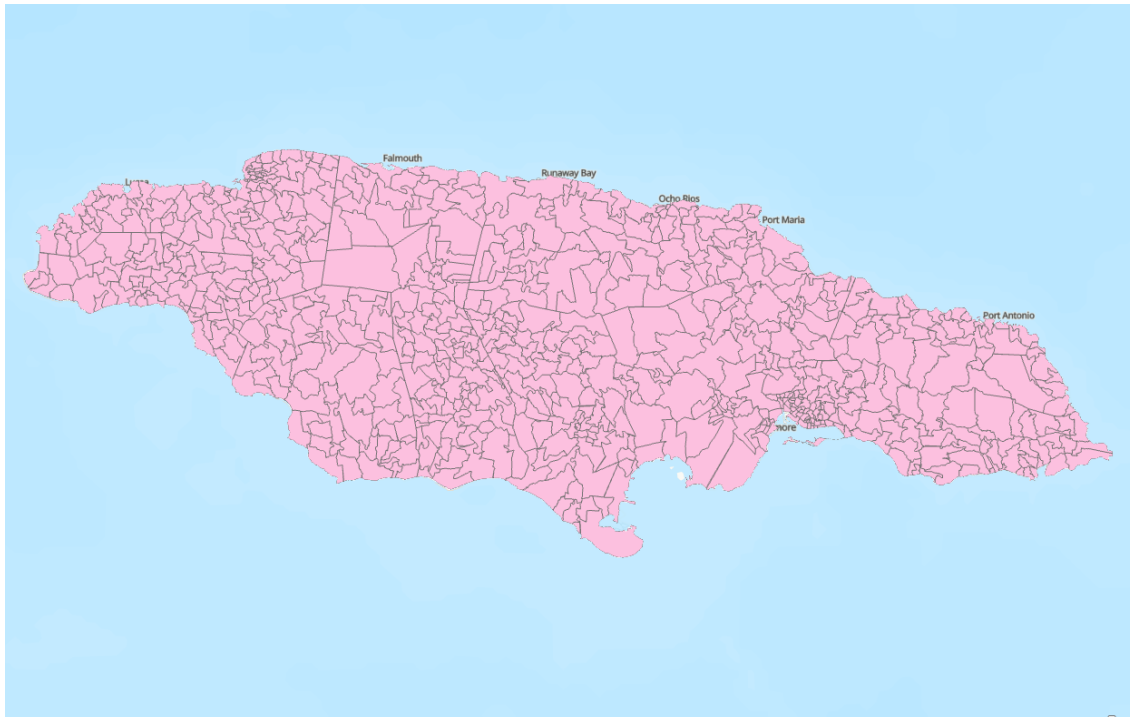


Figure 13: Jamaica Community Boundaries 2011 (STATIN Geographic Services Unit)

Table 22: Development Area Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Population	84	32,107.25	37,109.81	1,329	192,044
Area	84	130.52	88.58	3.56	446.33
Population.Density	84	610.98	1,429.67	42.12	8,410.05

Notes: Area is measured in square kilometers. Population is based on the 2011 Census.

8.7 Appendix 7: Additional Household and Labor Force Statistics

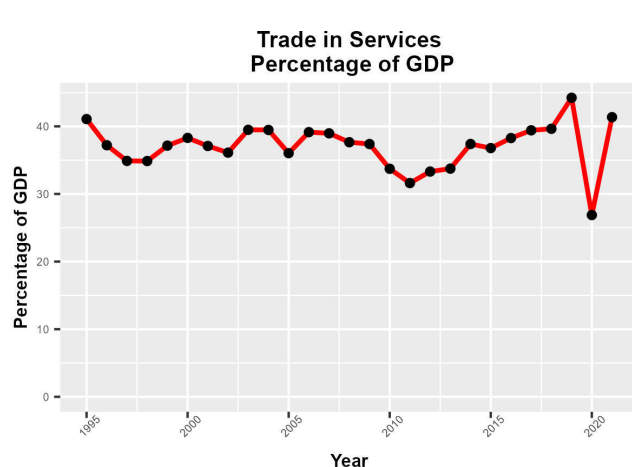


Figure 14: Services Sector Share
Source: World Bank

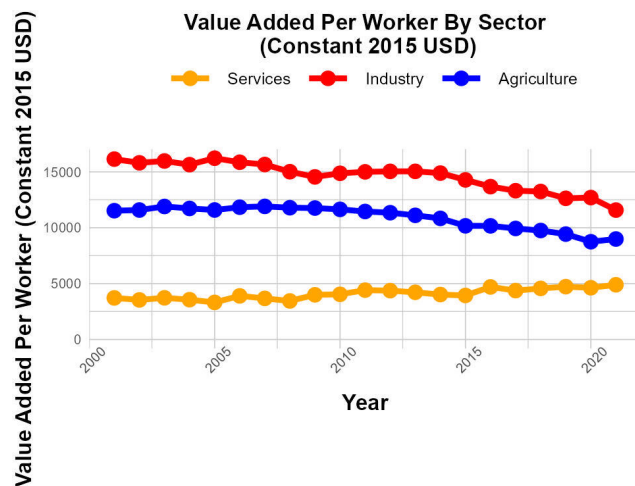


Figure 15: Multi-Sector Value Added
Source: World Bank

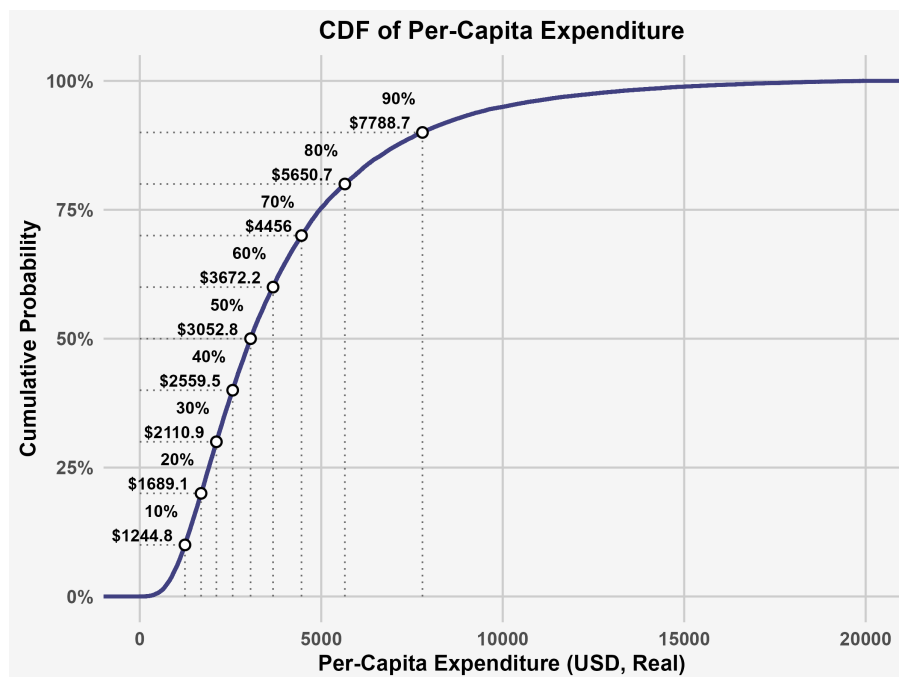


Figure 16: Real Per-Capita Expenditure Distribution of Jamaica Households
Source: Jamaica Survey of Living Conditions

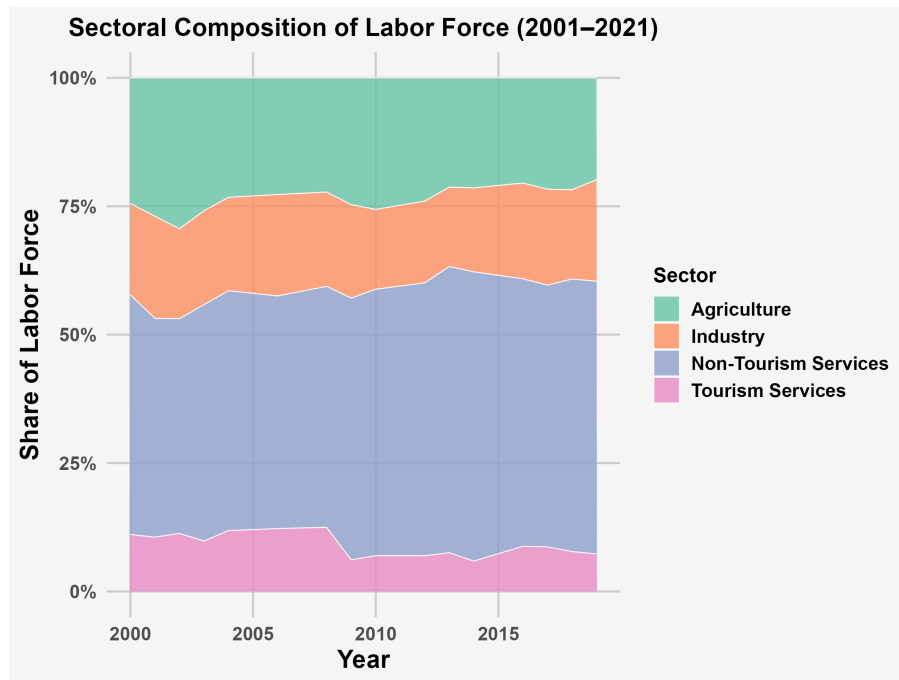


Figure 17: Sector Shares of Jamaican Labor Force
Source: Jamaica Survey of Living Conditions

Panel A: Agriculture vs Tourism Sector

Table 24: Agriculture vs. Tourism Services Comparison

	Agriculture		Tourism Services		Difference
	Mean	SD	Mean	SD	T-Stat
Per-Capita Consumption	2965.58	2372.50	3548.34	3046.32	-19.48***
Per-Capita Total Expenditure	3186.53	2767.628	3915.44	3878.217	-18.82***
Per-Capita Food Expenditure	1646.33	1346.511	1655.67	1295.775	-13.87***
Per-Capita Non-Food Expenditure	1319.75	1346.134	1893.32	2194.854	-18.72***
Per Capita Non Consumption Expenditure	341.91	897.240	449.24	1523.801	-8.50***
Non-Food Share of Consumption Expenditure	0.43	0.143	0.51	0.136	-22.66***
Non-Food Share of Tot. Expenditure	0.42	0.140	0.48	0.132	-17.70***
Consumption Share of Tot. Expenditure	0.96	0.083	0.95	0.091	11.81***
Years of Schooling	10.27	3.765	12.86	4.432	-17.95***
Household Decile	5.25	2.818	6.04	2.680	-27.06***
Male HH Head	0.54	0.499	0.61	0.488	0.23
Female HH Head	0.15	0.352	0.34	0.474	-15.37***
Single Male	0.30	0.458	0.04	0.196	18.22***
Single Female	0.02	0.141	0.01	0.096	-3.39***
Observations	6621		2862		9483

Panel B: Manufacturing vs. Tourism

Table 25: Manufacturing vs. Tourism Services Comparison

	Manufacturing		Tourism Services		Difference
	Mean	SD	Mean	SD	T-Stat
Per-Capita Consumption	3940.06	3235.83	3548.34	3046.32	-5.54***
Per-Capita Total Expenditure	4422.46	4229.719	3915.44	3878.217	-5.01***
Per-Capita Food Expenditure	1902.13	1430.458	1655.67	1295.775	-5.07***
Per-Capita Non-Food Expenditure	2037.93	2244.189	1893.32	2194.854	-4.71***
Per Capita Non Consumption Expenditure	627.78	1744.541	449.24	1523.801	-1.48
Non-Food Share of Consumption Expenditure	0.50	0.143	0.51	0.136	-2.40*
Non-Food Share of Tot. Expenditure	0.46	0.138	0.48	0.132	-1.94
Consumption Share of Tot. Expenditure	0.94	0.100	0.95	0.091	1.25
Years of Schooling	12.35	2.790	12.86	4.432	-2.69**
Household Decile	6.43	2.732	6.04	2.680	-5.27***
Male HH Head	0.52	0.500	0.61	0.488	-0.80
Female HH Head	0.25	0.433	0.34	0.474	-2.83**
Single Male	0.20	0.403	0.04	0.196	4.91***
Single Female	0.02	0.156	0.01	0.096	-1.55
Observations	1330		2862		4192

Notes: All statistics are weighted by household size. Panel A reports means, medians, and standard deviations for the full sample. Panel B compares urban and rural households using t-tests with unequal variances. *** p<0.01, ** p<0.05, * p<0.1.